Measured Evidence Summary

Dataset: CIFAR-10 official test split (10,000 samples)

Checkpoint: 9,555/10,000 = 95.55%

Exported integer arithmetic: 9,542/10,000 = 95.42%

Prediction disagreements: 257

Model MAC equivalents: 3,732,164,352

Tile commands: 104,256

Hardware parameter archive: 9,385,704 bytes, SHA-256 b02c5df519eaecffb228f4c8be21babddc01761f26210bb8dfcca3318ed0d381

Power pages use Vivado report data only; no physical-board value is inferred.

Full PYNQ routed setup/hold slack: +0.005 / +0.050 ns

Clock domains: accelerator 200 MHz, DMA 125 MHz, control 100 MHz

Full-system SAIF estimate: 1.937 W total, 1.785 W dynamic, confidence Medium, nets matched 7% (9164/123250)

Complete Exported-Graph Trace Validation

Result: PASS

Official CIFAR-10 fixture: outputs/cifar10_shards/cifar10_test_000.npz:0

Label / predicted class: 3 / 3

Complete graph commands: 1,345,393

Input / output stream beats: 41,408,322 / 13,105,354

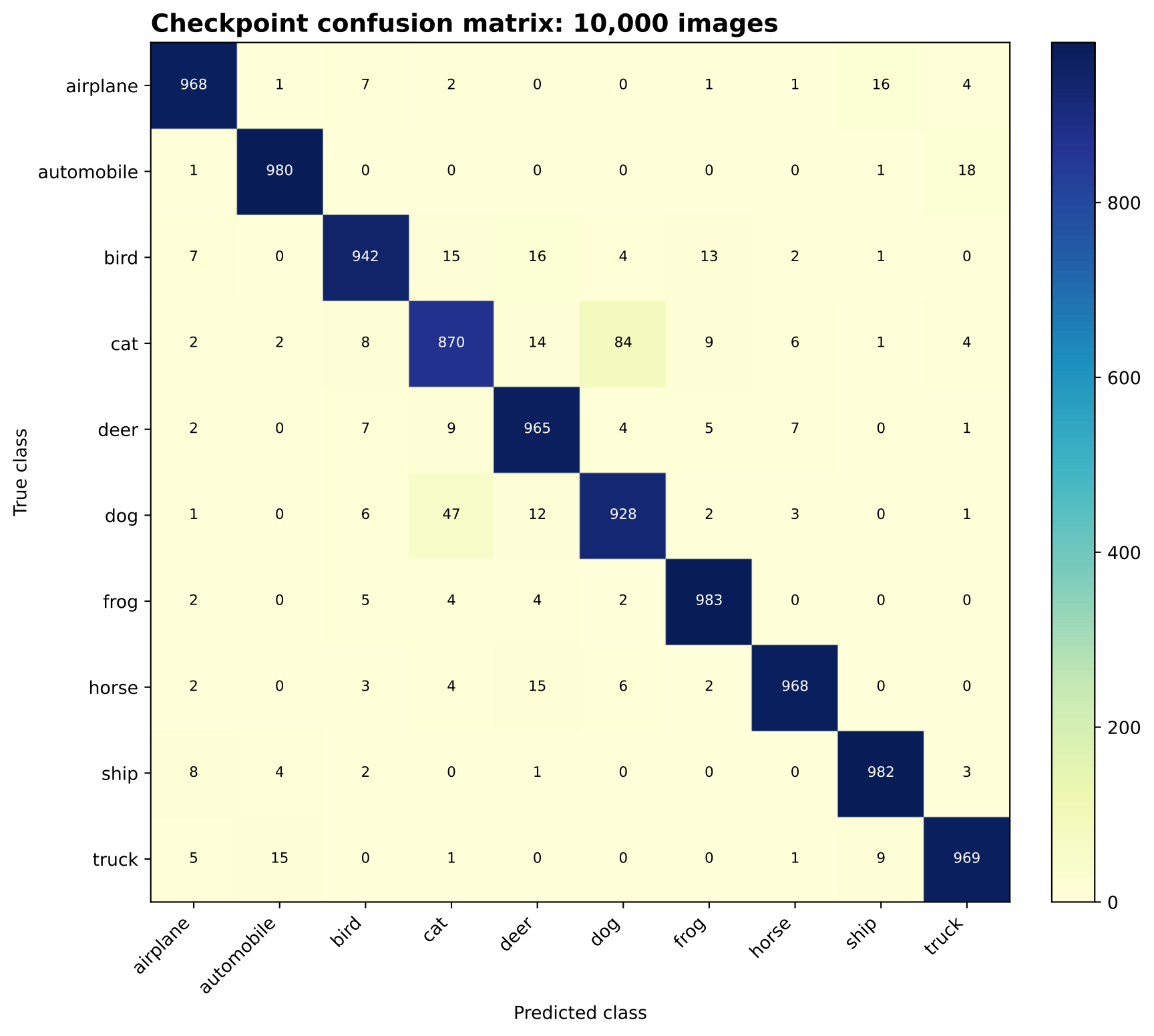
Maximum logit delta from archived 10,000-image evaluation: 9.596e-08

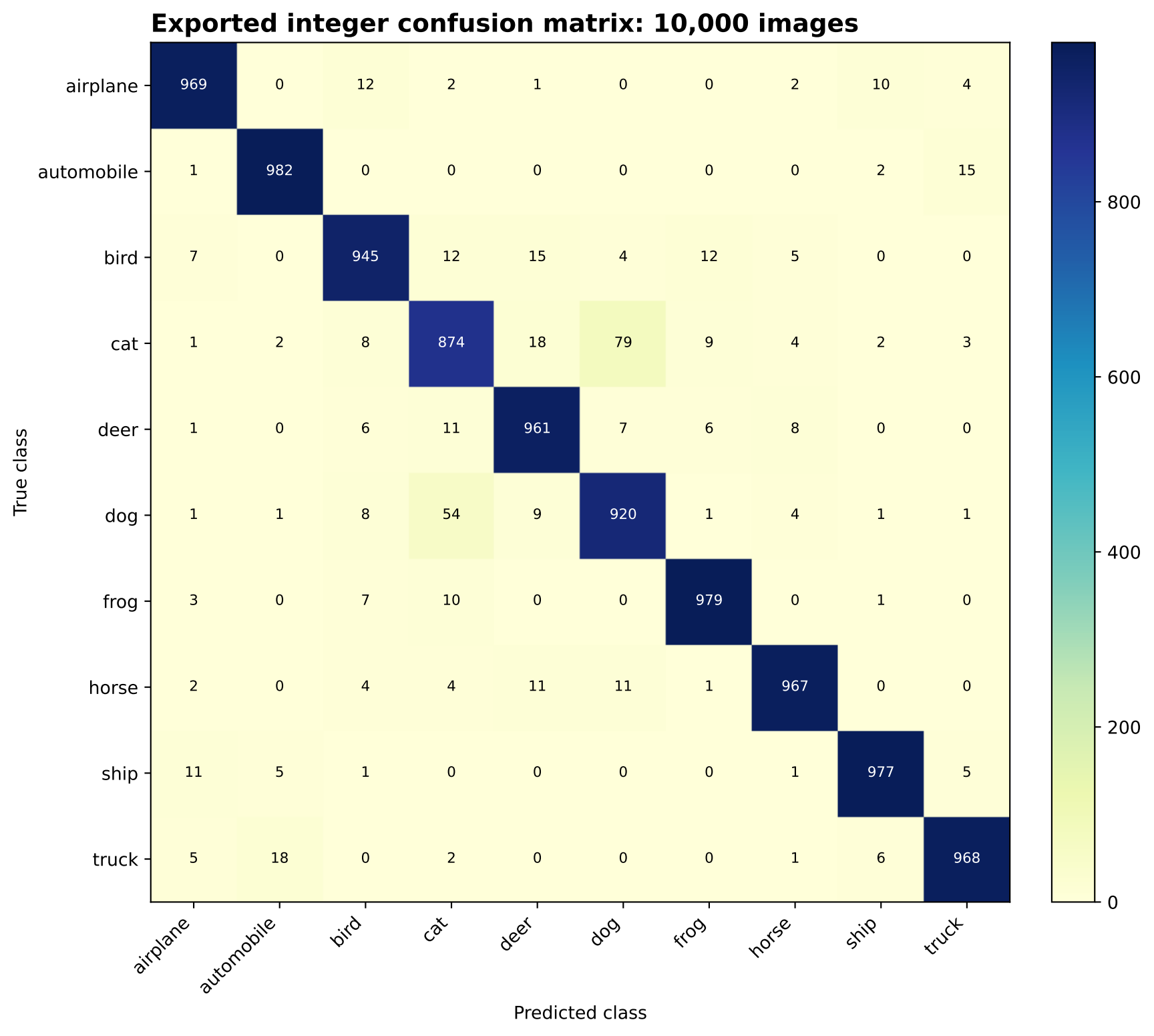
Parameter bytes / SHA-256: 9,385,704 / b02c5df519eaecffb228f4c8be21babddc01761f26210bb8dfcca3318ed0d381

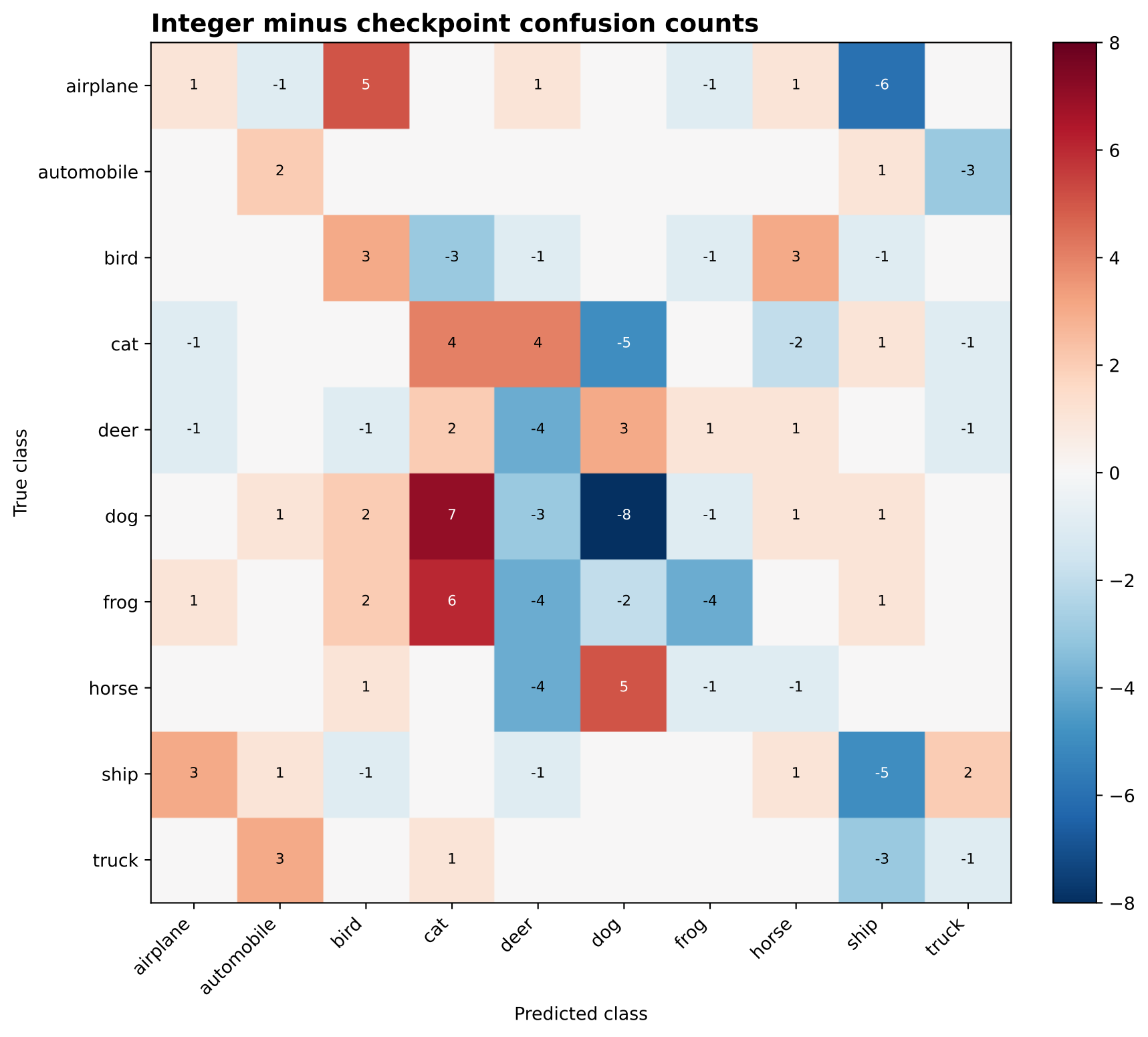
Host trace elapsed time: 197.466 s

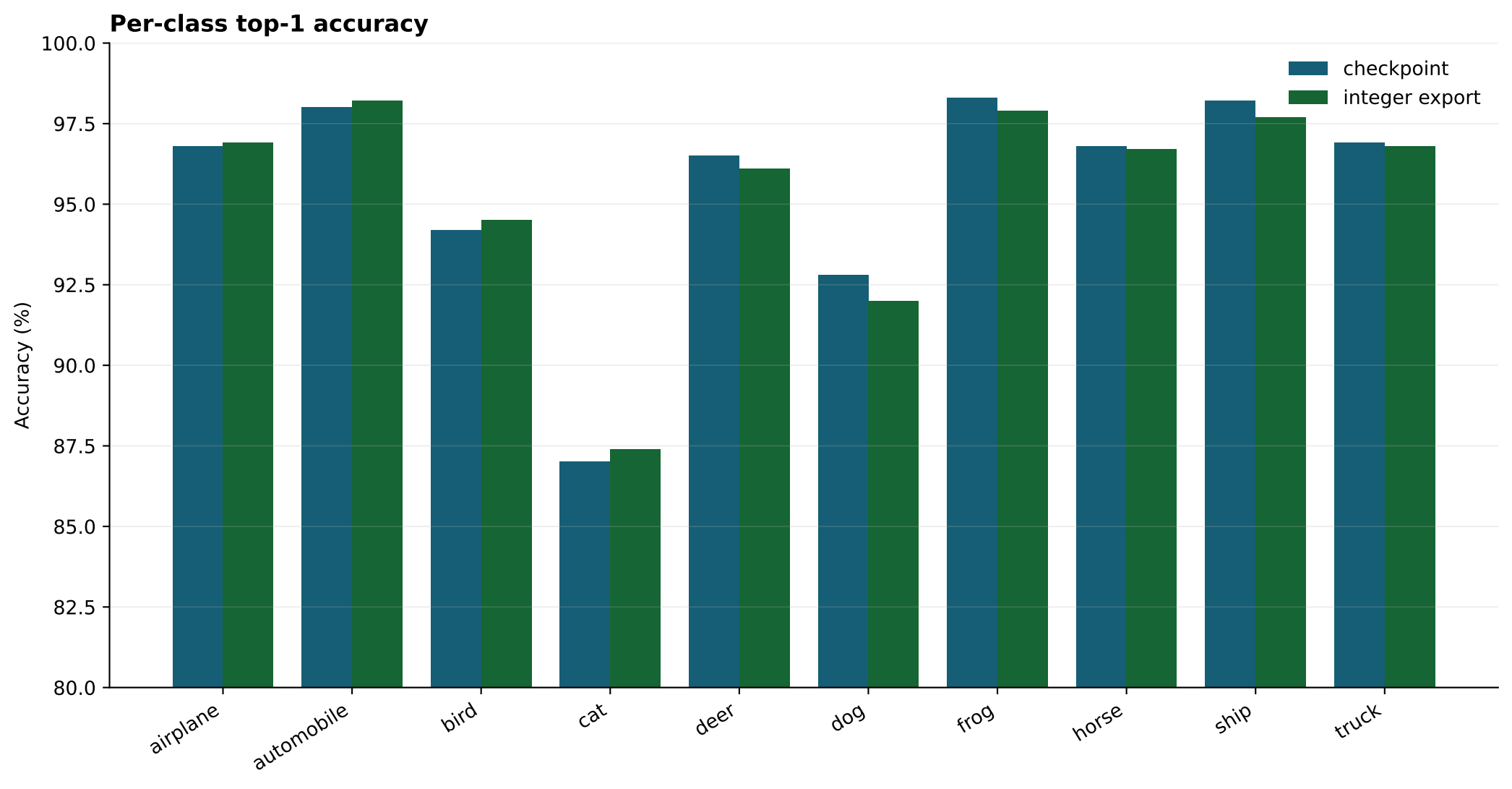
Final PYNQ Deliverable Integrity

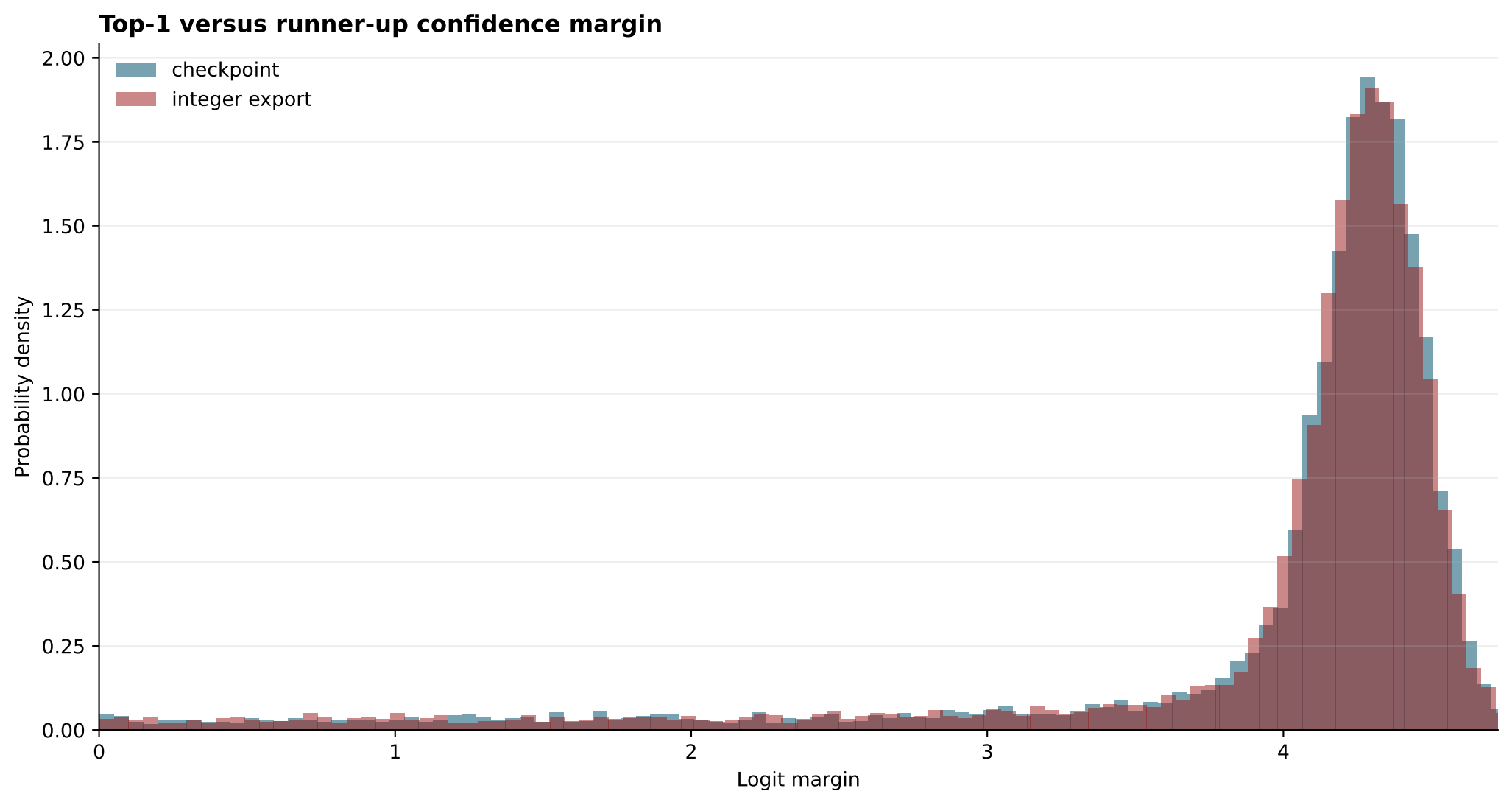
```
neuraedge_pynq_full.bit: 4,045,769 bytes
  SHA-256 d9bcac6224efd44da8aa2f8a0e674f5e66a67294caac460d6cfc8086d5a341de
neuraedge_pynq_full.hwh: 384,395 bytes
  SHA-256 79395d99357fa798ecd31b0c9229500dce7e7aada8185c3576b744fd4485712f
neuraedge_pynq_full.xsa: 2,814,632 bytes
  SHA-256 8d06b8a0b6e82105992cbf05fd1f0d3cfac8f5316d682f976f88221b0583bd64
neuraedge_pynq_full_routed.dcp: 32,256,514 bytes
  SHA-256 750e7185560d3187c7576f84b722918d3c8d3da21738d417b8fa59fc9ca857a1
```

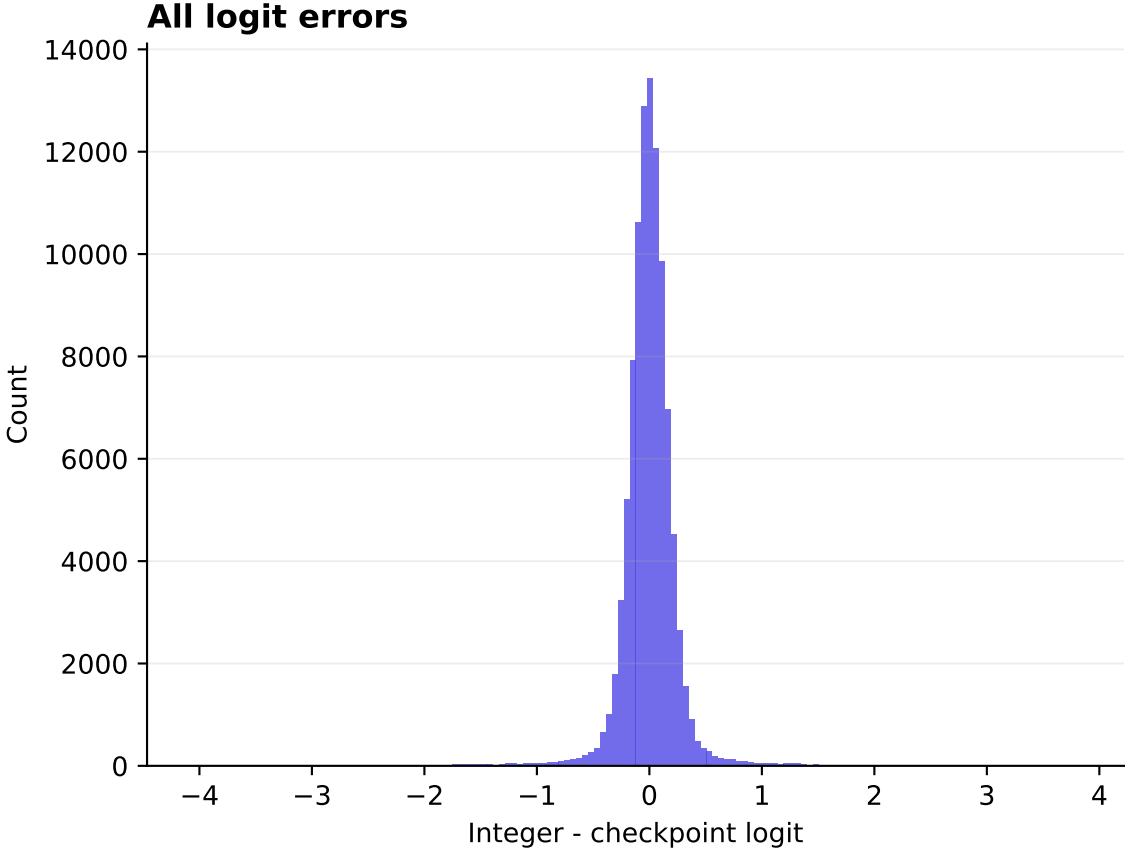

Checkpoint confusion matrix: 10,000 images
True class
Predicted class
airplane
automobile
bird
cat
deer
dog
frog
horse
ship
truck
968 1 7 2 0 0 1 1 16 4
1 980 0 0 0 0 0 0 1 18
7 0 942 15 16 4 13 2 1 0
2 2 8 870 14 84 9 6 1 4
2 0 7 9 965 4 5 7 0 1
1 0 6 47 12 928 2 3 0 1
2 0 5 4 4 2 983 0 0 0
2 0 3 4 15 6 2 968 0 0
8 4 2 0 1 0 0 0 982 3
5 15 0 1 0 0 0 1 9 969
800
600
400
200
0

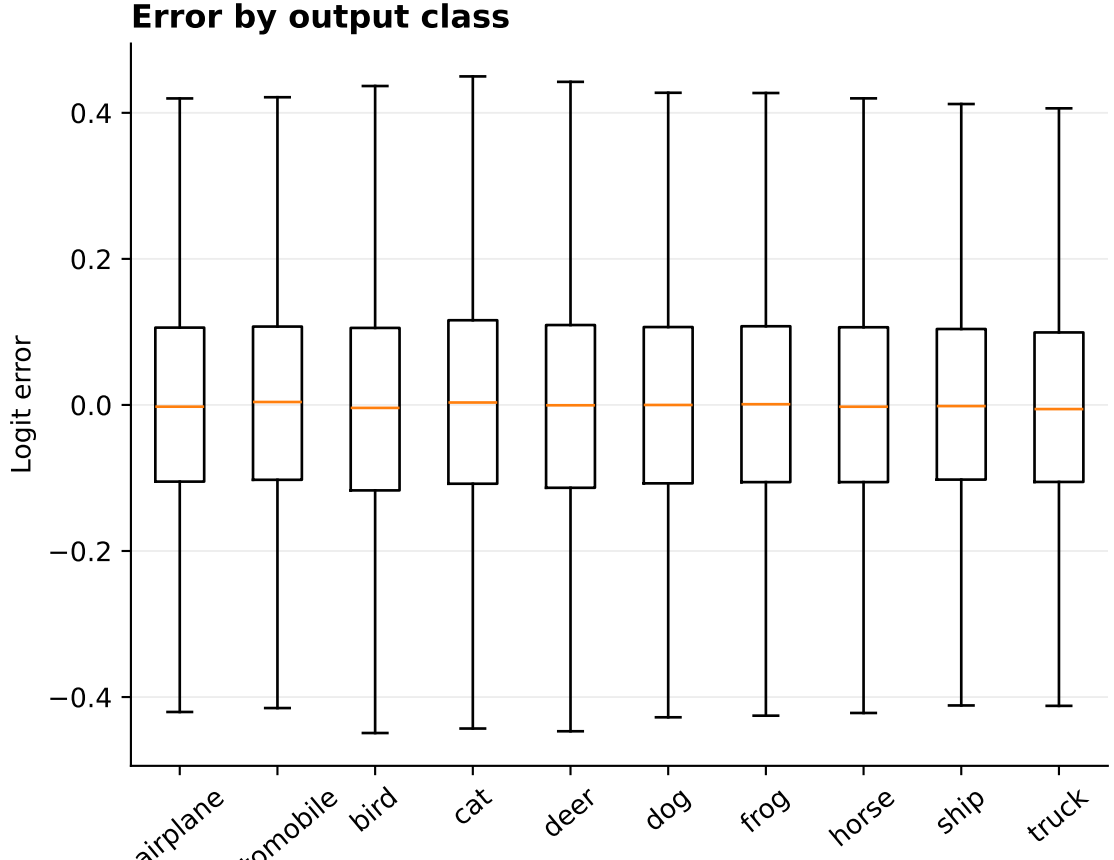

Exported integer confusion matrix: 10,000 images
True class
Predicted class
airplane
automobile
bird
cat
deer
dog
frog
horse
ship
truck
969 0 12 2 1 0 0 2 10 4
1 982 0 0 0 0 0 0 2 15
7 0 945 12 15 4 12 5 0 0
1 2 8 874 18 79 9 4 2 3
1 0 6 11 961 7 6 8 0 0
1 1 8 54 9 920 1 4 1 1
3 0 7 10 0 0 979 0 1 0
2 0 4 4 11 11 1 967 0 0
11 5 1 0 0 0 0 1 977 5
5 18 0 2 0 0 0 1 6 968
800
600
400
200
0

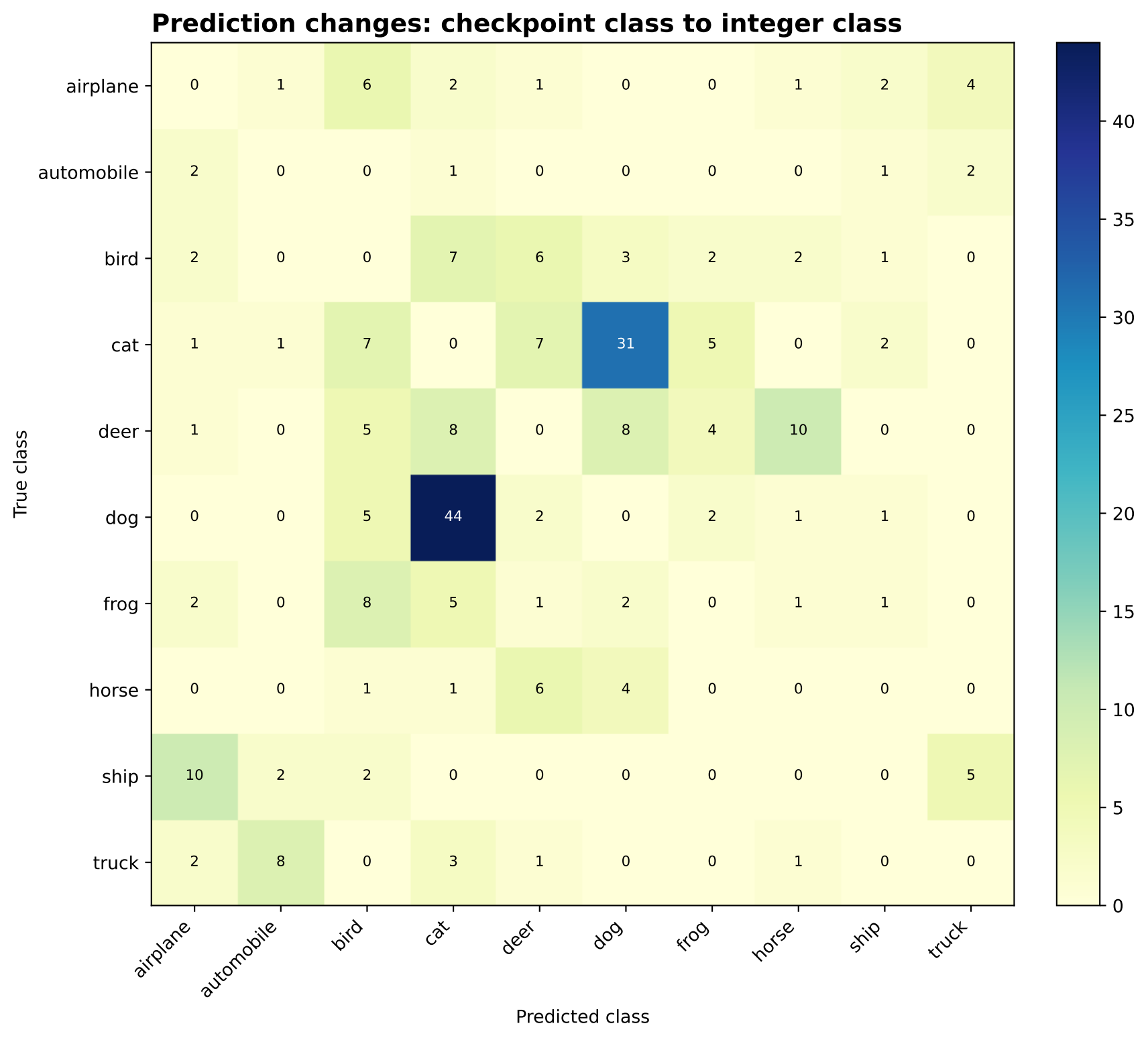

Integer minus checkpoint confusion counts
True class
Predicted class
airplane
automobile
bird
cat
deer
dog
frog
horse
ship
truck
1 -1 5 1 -1 1 -6
2 1 -3
3 -3 -1 -1 3 -1
-1 4 4 -5 -2 1 -1
-1 -1 2 -4 3 1 1 -1
1 2 7 -3 -8 -1 1 1
1 2 6 -4 -2 -4 1
1 -4 5 -1 -1
3 1 -1 -1 1 -5 2
3 1 -3 -1
8
6
4
2
0
−2
−4
−6
−8

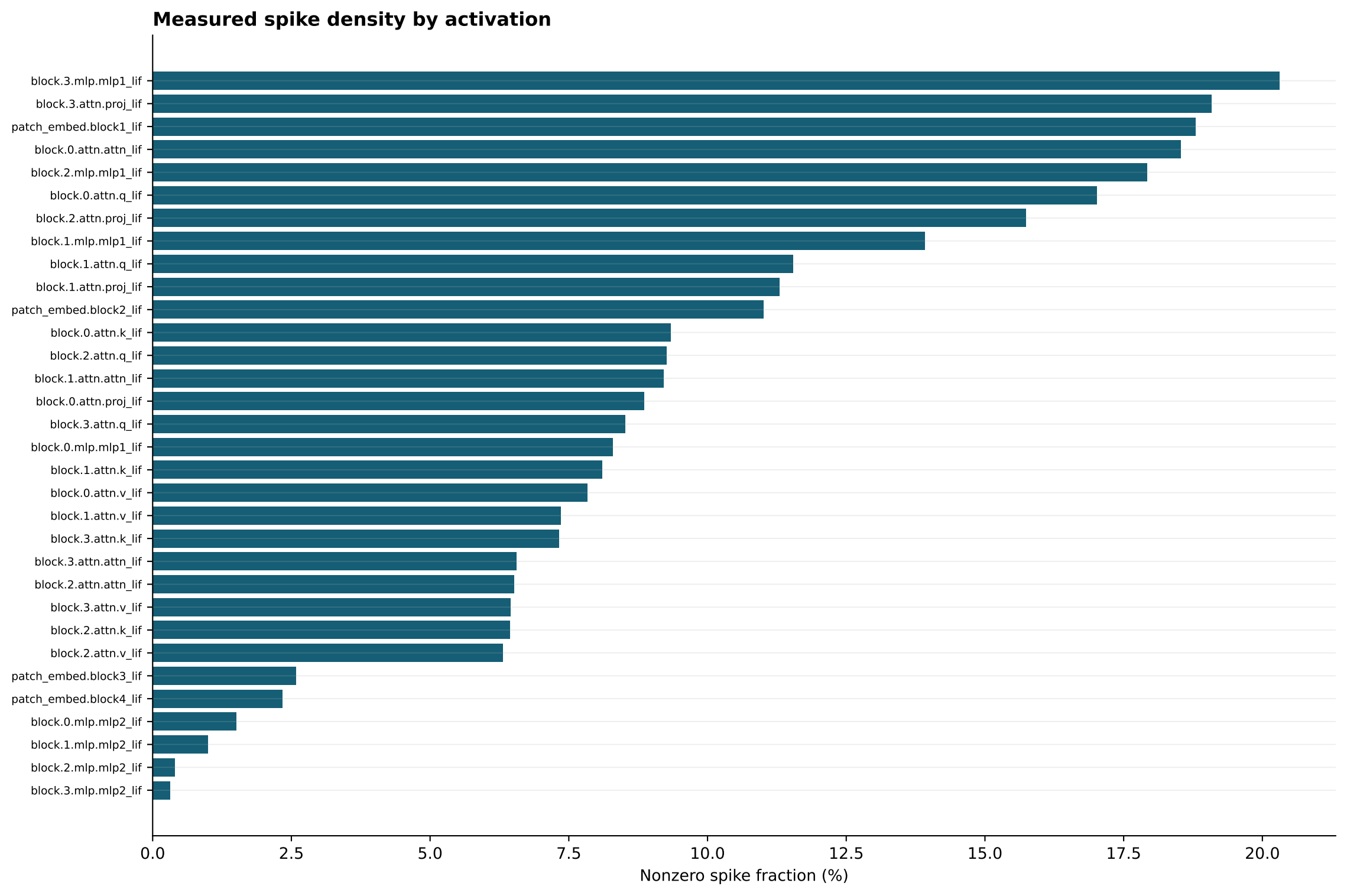

Per-class top-1 accuracy
checkpoint
integer export
Accuracy (%)
100.0
97.5
95.0
92.5
90.0
87.5
85.0
82.5
80.0
airplane
automobile
bird
cat
deer
dog
frog
horse
ship
truck

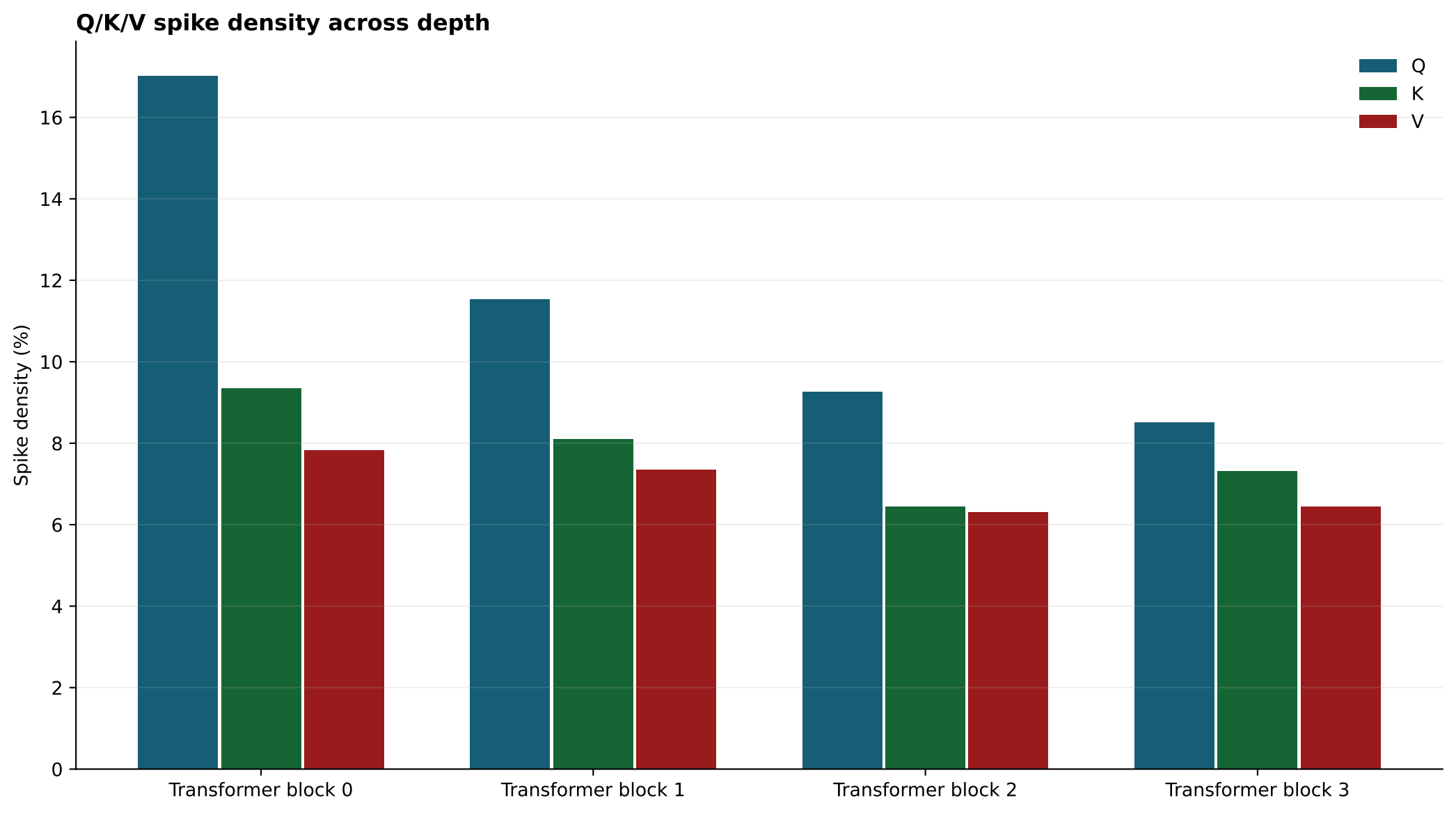

Top-1 versus runner-up confidence margin
checkpoint
integer export
Probability density
Logit margin
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
0
1
2
3
4

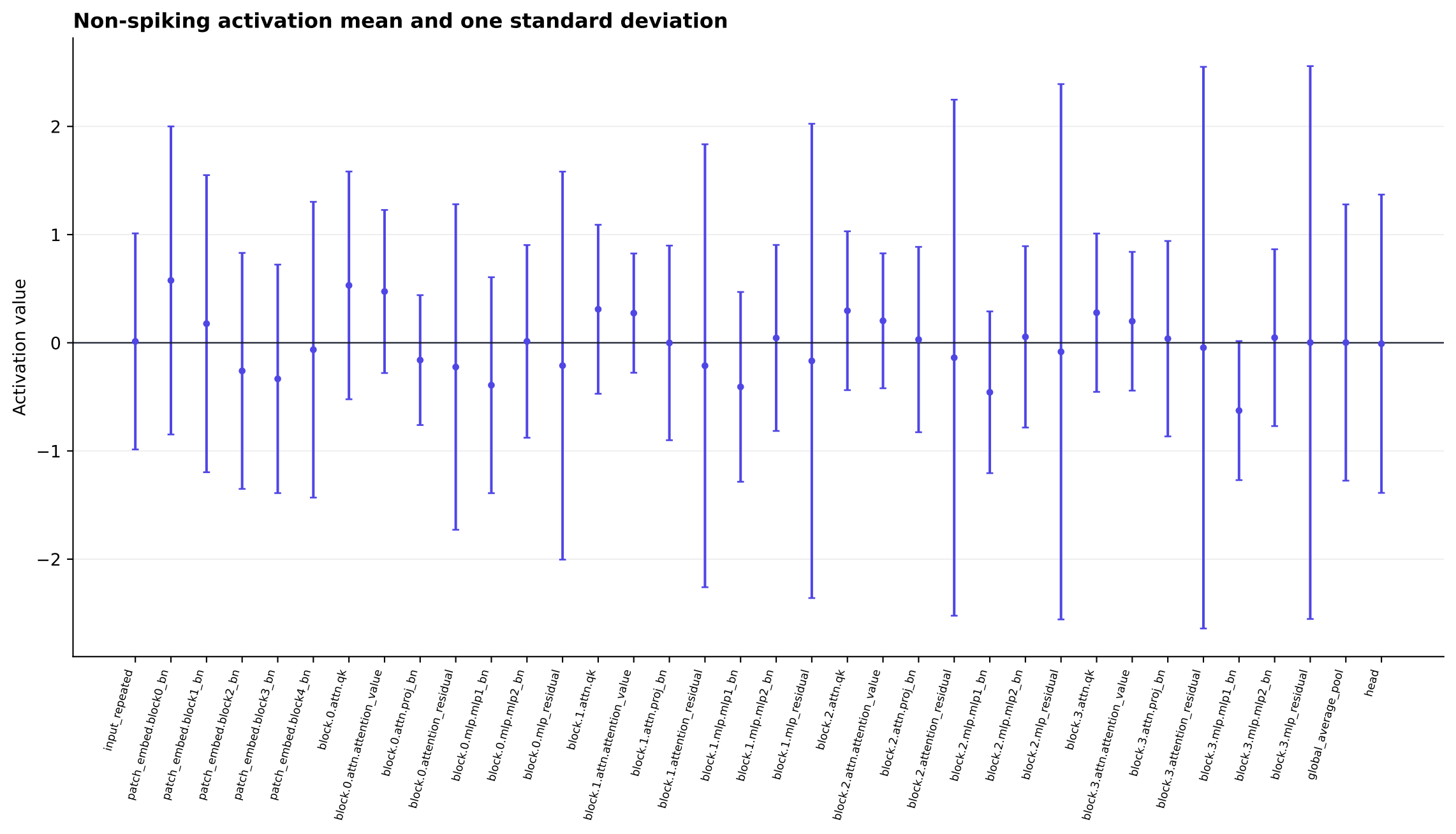

All logit errors
Count
Integer - checkpoint logit
0
2000
4000
6000
8000
10000
12000
14000
−4
−3
−2
−1
0
1
2
3
4

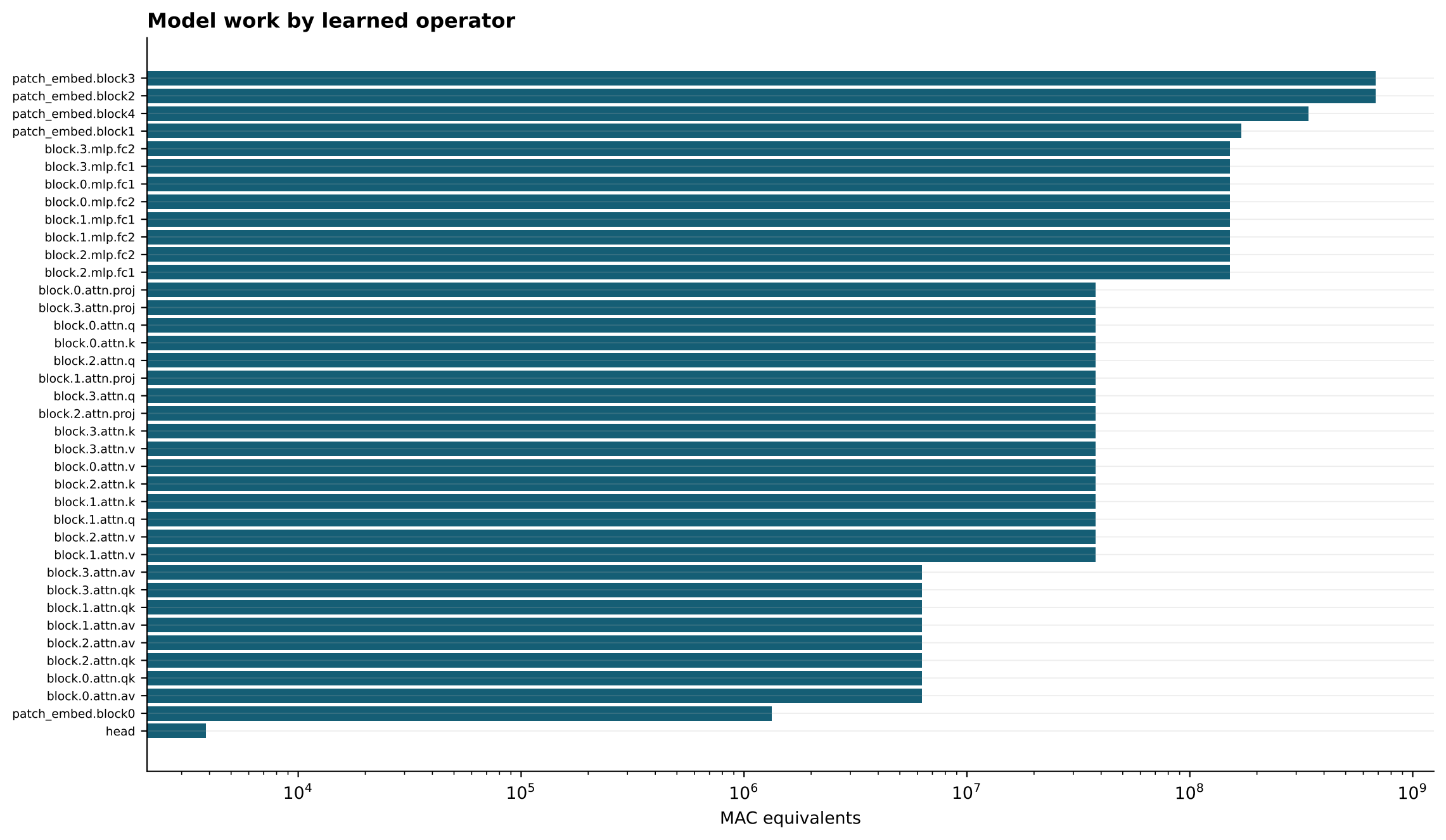

Error by output class
Logit error
0.4
0.2
0.0
−0.2
−0.4
airplane
automobile
bird
cat
deer
dog
frog
horse
ship
truck

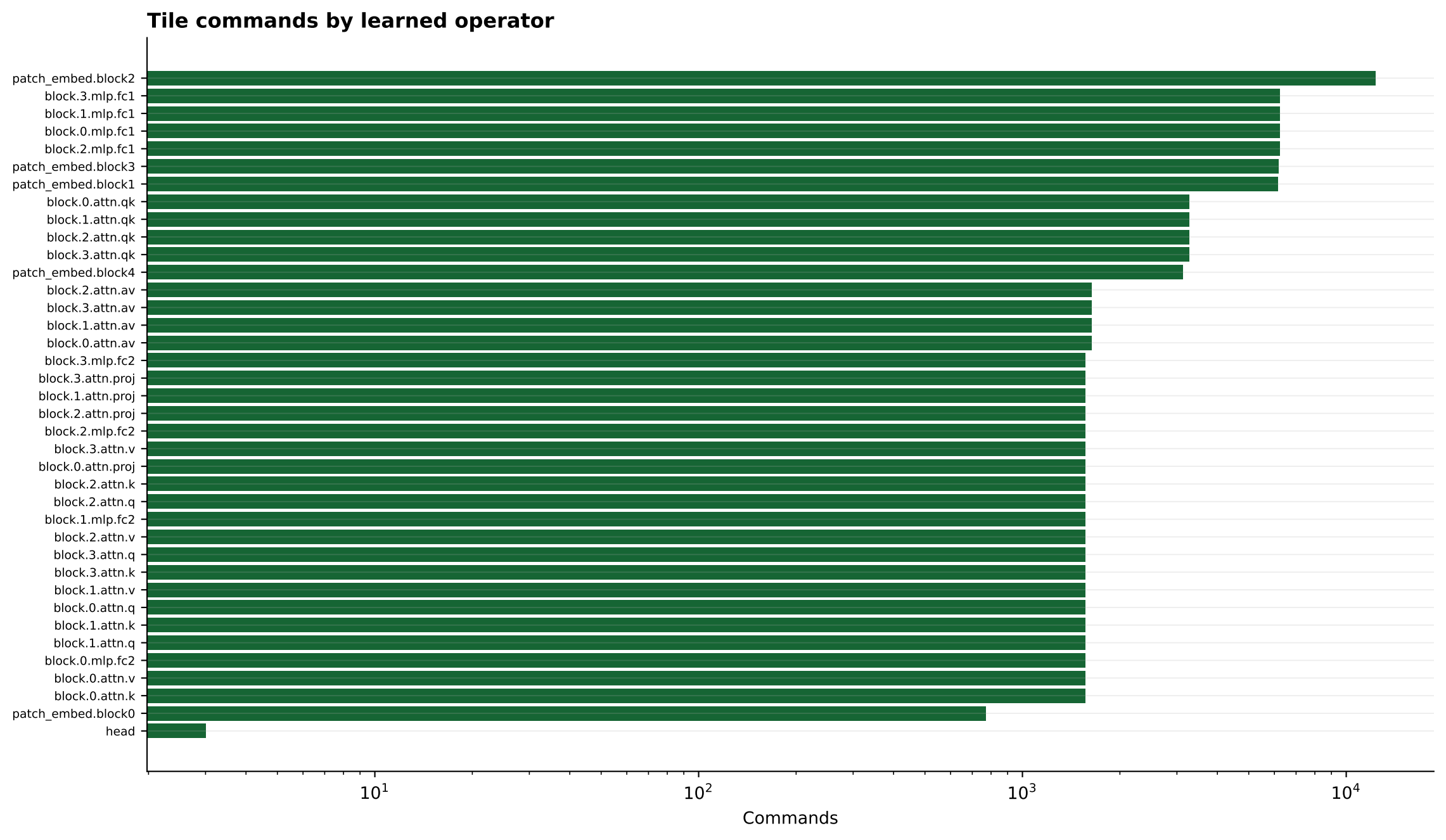

Prediction changes: checkpoint class to integer class
True class
Predicted class
airplane
automobile
bird
cat
deer
dog
frog
horse
ship
truck
0 1 6 2 1 0 0 1 2 4
2 0 0 1 0 0 0 0 1 2
2 0 0 7 6 3 2 2 1 0
1 1 7 0 7 31 5 0 2 0
1 0 5 8 0 8 4 10 0 0
0 0 5 44 2 0 2 1 1 0
2 0 8 5 1 2 0 1 1 0
0 0 1 1 6 4 0 0 0 0
10 2 2 0 0 0 0 0 0 5
2 8 0 3 1 0 0 1 0 0
0
5
10
15
20
25
30
35
40

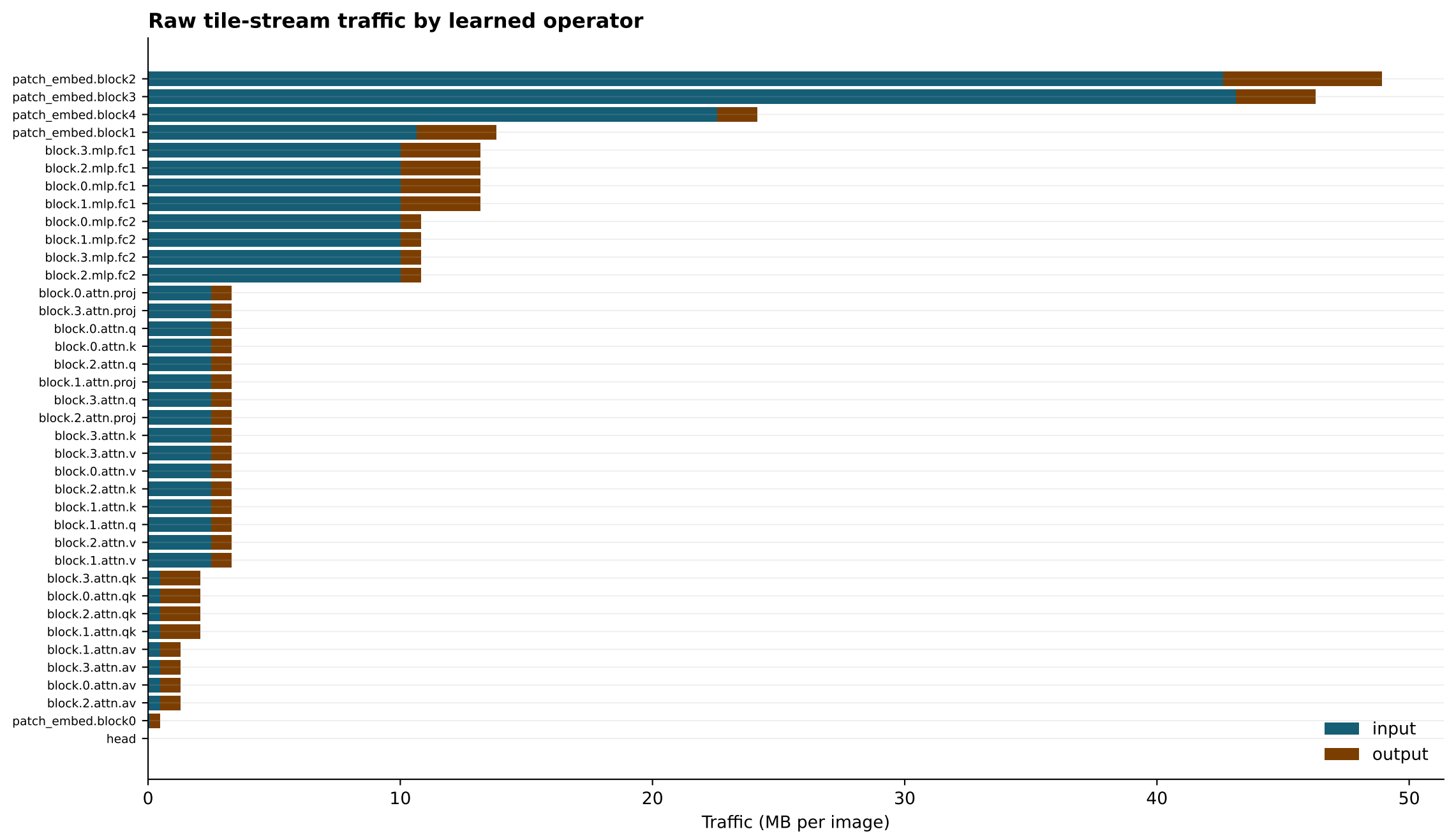

Measured spike density by activation
block.3.mlp.mlp1_lif
block.3.attn.proj_lif
patch_embed.block1_lif
block.0.attn.attn_lif
block.2.mlp.mlp1_lif
block.0.attn.q_lif
block.2.attn.proj_lif
block.1.mlp.mlp1_lif
block.1.attn.q_lif
block.1.attn.proj_lif
patch_embed.block2_lif
block.0.attn.k_lif
block.2.attn.q_lif
block.1.attn.attn_lif
block.0.attn.proj_lif
block.3.attn.q_lif
block.0.mlp.mlp1_lif
block.1.attn.k_lif
block.0.attn.v_lif
block.1.attn.v_lif
block.3.attn.k_lif
block.3.attn.attn_lif
block.2.attn.attn_lif
block.3.attn.v_lif
block.2.attn.k_lif
block.2.attn.v_lif
patch_embed.block3_lif
patch_embed.block4_lif
block.0.mlp.mlp2_lif
block.1.mlp.mlp2_lif
block.2.mlp.mlp2_lif
block.3.mlp.mlp2_lif
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Nonzero spike fraction (%)

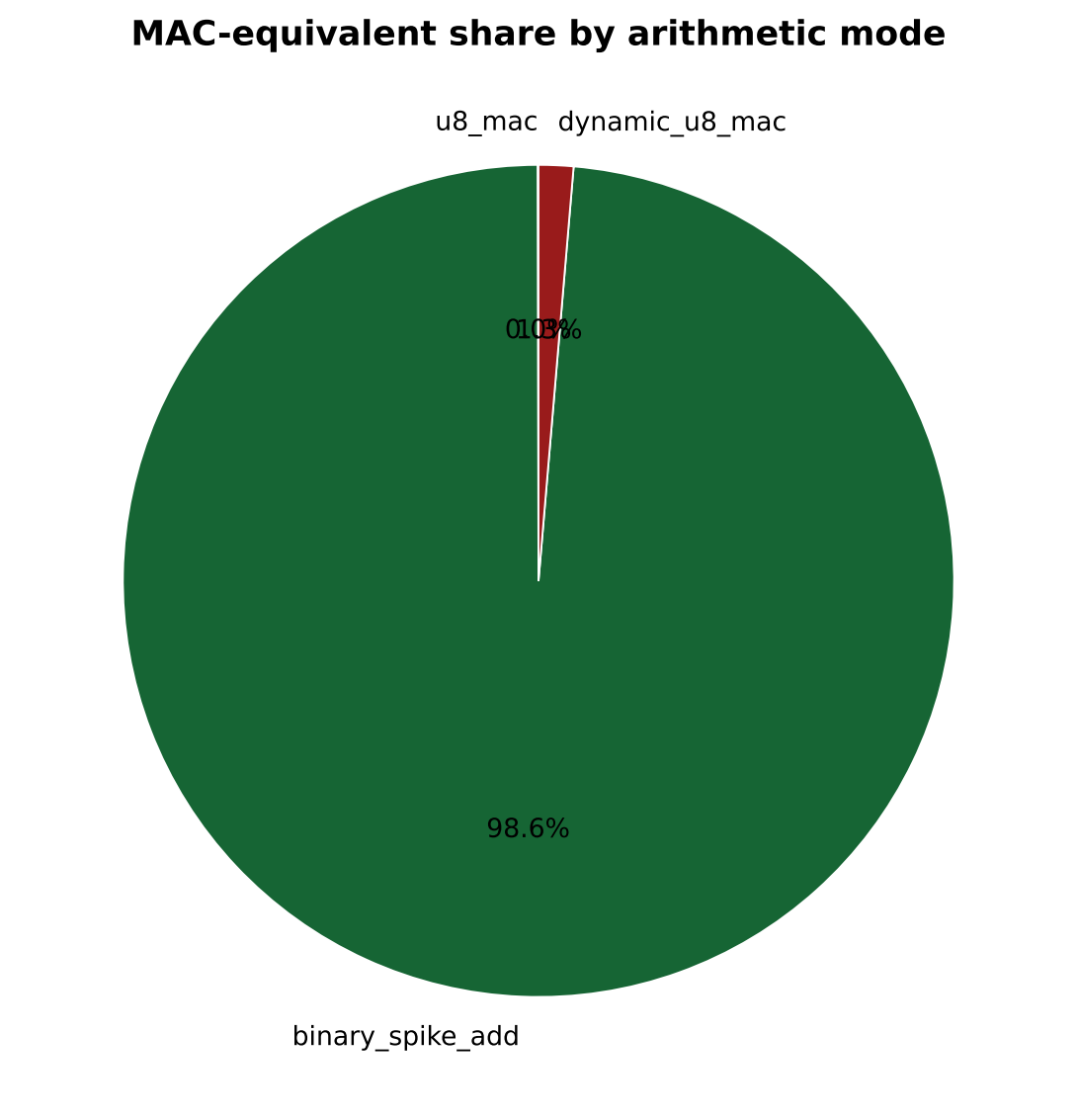

Q/K/V spike density across depth
Q
K
V
Spike density (%)
0
2
4
6
8
10
12
14
16
Transformer block 0
Transformer block 1
Transformer block 2
Transformer block 3

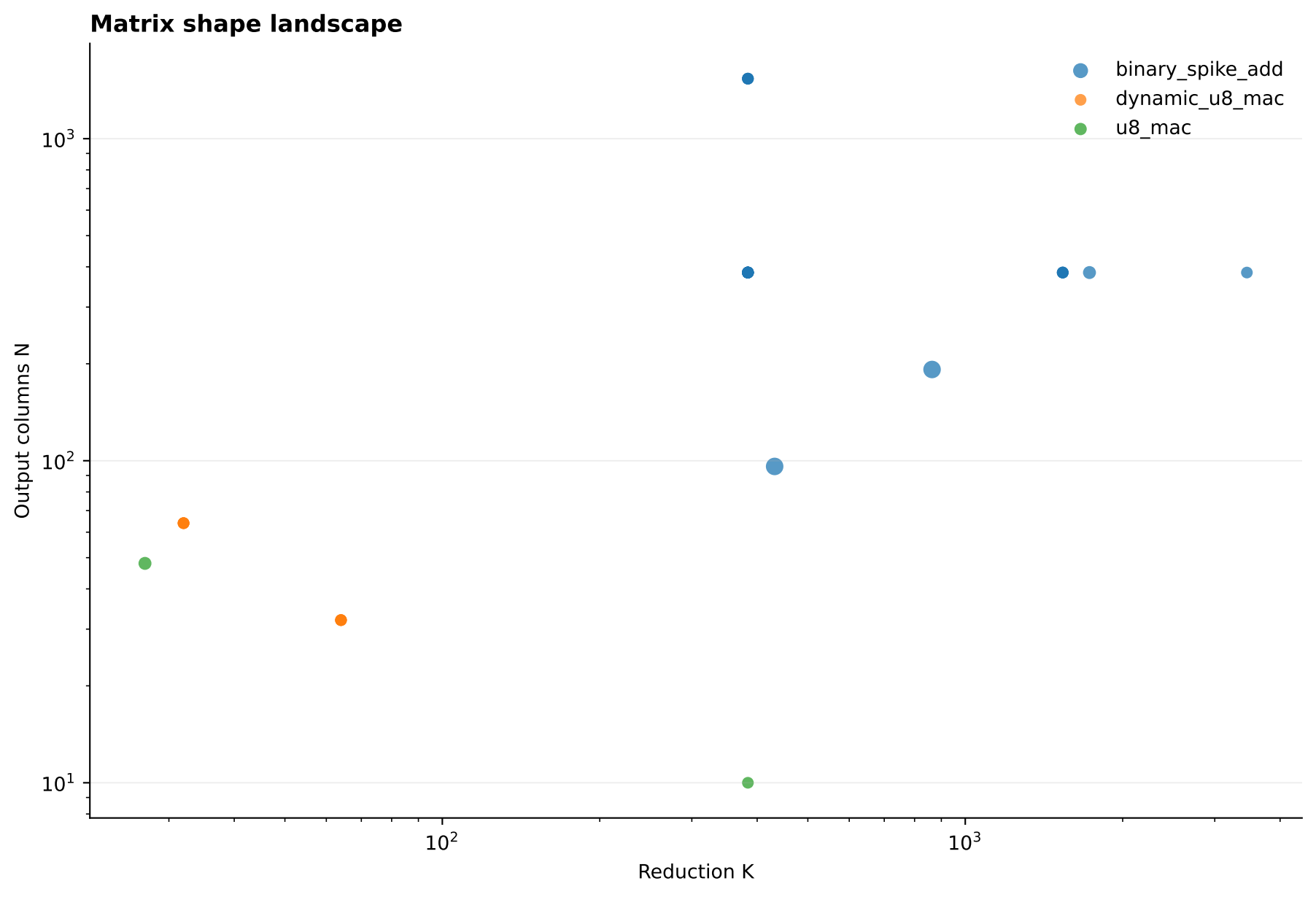

Non-spiking activation mean and one standard deviation
Activation value
2
1
0
−1
−2
input_repeated
patch_embed.block0_bn
patch_embed.block1_bn
patch_embed.block2_bn
patch_embed.block3_bn
patch_embed.block4_bn
block.0.attn.qk
block.0.attn.attention_value
block.0.attn.proj_bn
block.0.attention_residual
block.0.mlp.mlp1_bn
block.0.mlp.mlp2_bn
block.0.mlp_residual
block.1.attn.qk
block.1.attn.attention_value
block.1.attn.proj_bn
block.1.attention_residual
block.1.mlp.mlp1_bn
block.1.mlp.mlp2_bn
block.1.mlp_residual
block.2.attn.qk
block.2.attn.attention_value
block.2.attn.proj_bn
block.2.attention_residual
block.2.mlp.mlp1_bn
block.2.mlp.mlp2_bn
block.2.mlp_residual
block.3.attn.qk
block.3.attn.attention_value
block.3.attn.proj_bn
block.3.attention_residual
block.3.mlp.mlp1_bn
block.3.mlp.mlp2_bn
block.3.mlp_residual
global_average_pool
head

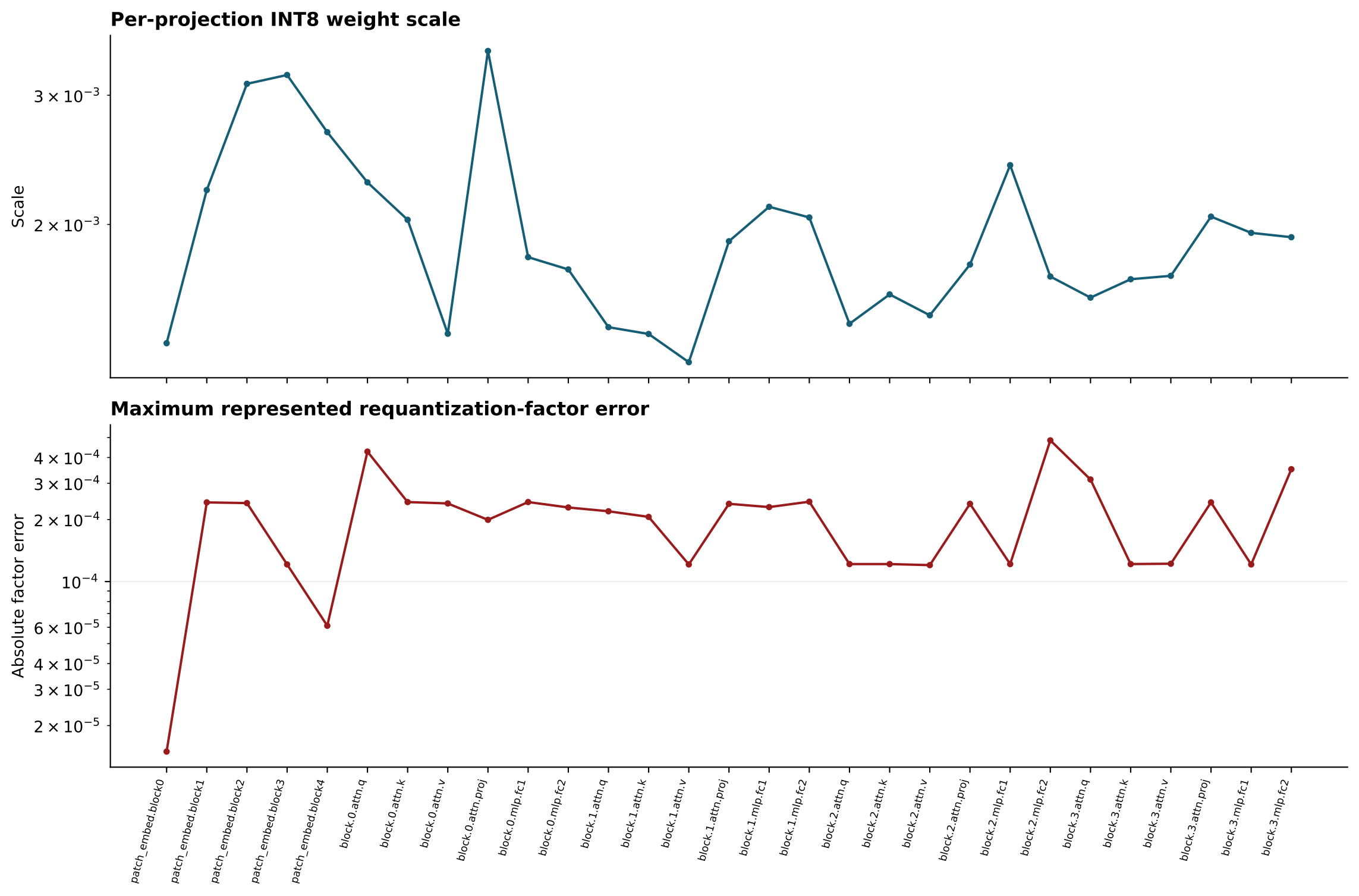

Model work by learned operator
patch_embed.block3
patch_embed.block2
patch_embed.block4
patch_embed.block1
block.3.mlp.fc2
block.3.mlp.fc1
block.0.mlp.fc1
block.0.mlp.fc2
block.1.mlp.fc1
block.1.mlp.fc2
block.2.mlp.fc2
block.2.mlp.fc1
block.0.attn.proj
block.3.attn.proj
block.0.attn.q
block.0.attn.k
block.2.attn.q
block.1.attn.proj
block.3.attn.q
block.2.attn.proj
block.3.attn.k
block.3.attn.v
block.0.attn.v
block.2.attn.k
block.1.attn.k
block.1.attn.q
block.2.attn.v
block.1.attn.v
block.3.attn.av
block.3.attn.qk
block.1.attn.qk
block.1.attn.av
block.2.attn.av
block.2.attn.qk
block.0.attn.qk
block.0.attn.av
patch_embed.block0
head
10^4
10^5
10^6
10^7
10^8
10^9
MAC equivalents

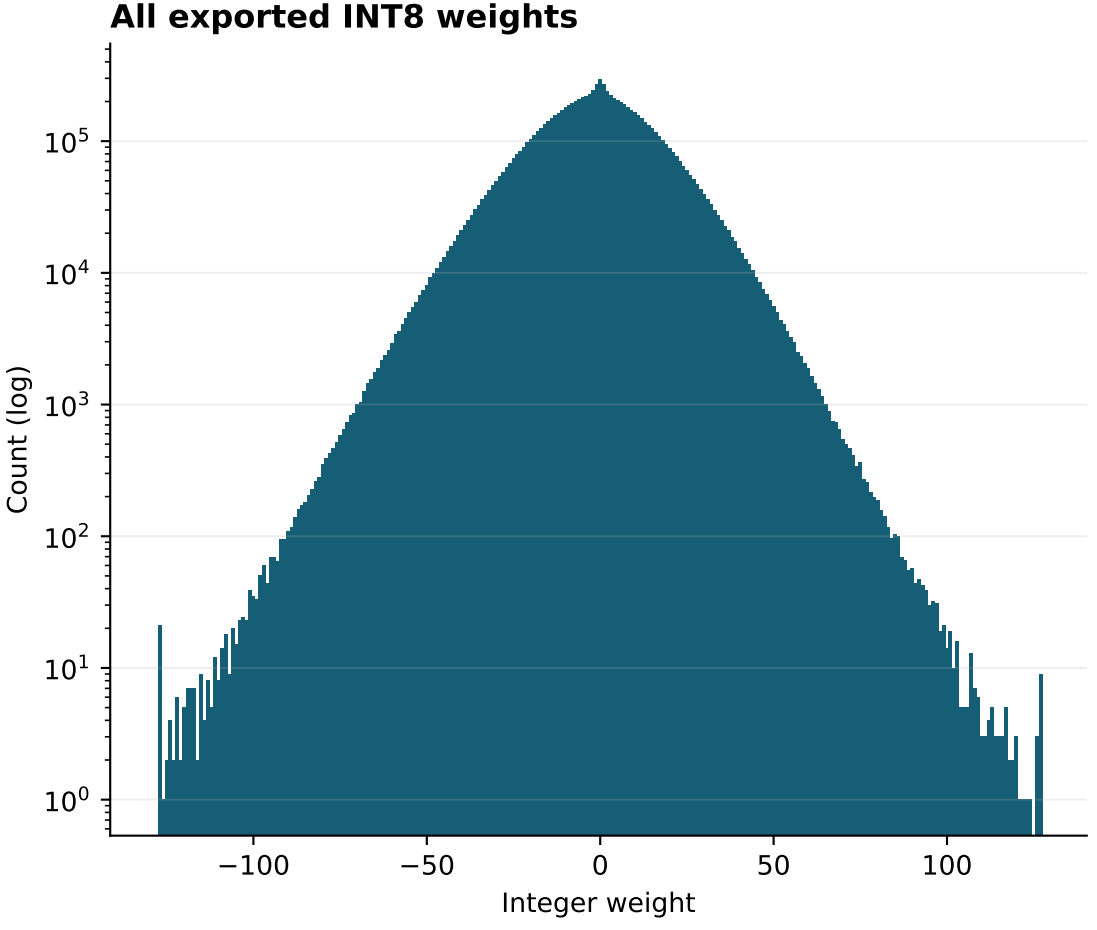

Tile commands by learned operator
patch_embed.block2
block.3.mlp.fc1
block.1.mlp.fc1
block.0.mlp.fc1
block.2.mlp.fc1
patch_embed.block3
patch_embed.block1
block.0.attn.qk
block.1.attn.qk
block.2.attn.qk
block.3.attn.qk
patch_embed.block4
block.2.attn.av
block.3.attn.av
block.1.attn.av
block.0.attn.av
block.3.mlp.fc2
block.3.attn.proj
block.1.attn.proj
block.2.attn.proj
block.2.mlp.fc2
block.3.attn.v
block.0.attn.proj
block.2.attn.k
block.2.attn.q
block.1.mlp.fc2
block.2.attn.v
block.3.attn.q
block.3.attn.k
block.1.attn.v
block.0.attn.q
block.1.attn.k
block.1.attn.q
block.0.mlp.fc2
block.0.attn.v
block.0.attn.k
patch_embed.block0
head
10^1
10^2
10^3
10^4
Commands

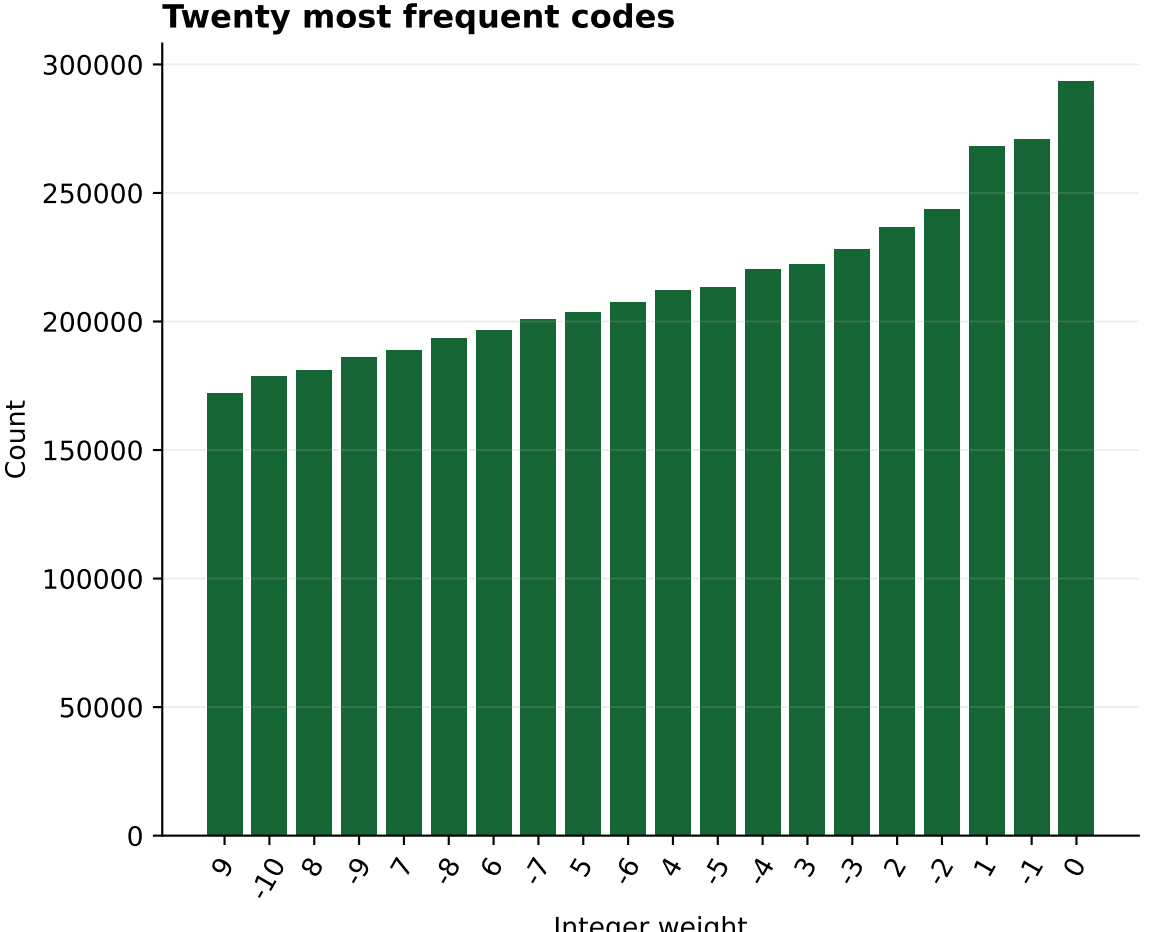

Raw tile-stream traffic by learned operator
patch_embed.block2
patch_embed.block3
patch_embed.block4
patch_embed.block1
block.3.mlp.fc1
block.2.mlp.fc1
block.0.mlp.fc1
block.1.mlp.fc1
block.0.mlp.fc2
block.1.mlp.fc2
block.3.mlp.fc2
block.2.mlp.fc2
block.0.attn.proj
block.3.attn.proj
block.0.attn.q
block.0.attn.k
block.2.attn.q
block.1.attn.proj
block.3.attn.q
block.2.attn.proj
block.3.attn.k
block.3.attn.v
block.0.attn.v
block.2.attn.k
block.1.attn.k
block.1.attn.q
block.2.attn.v
block.1.attn.v
block.3.attn.qk
block.0.attn.qk
block.2.attn.qk
block.1.attn.qk
block.1.attn.av
block.3.attn.av
block.0.attn.av
block.2.attn.av
patch_embed.block0
head
input
output
0
10
20
30
40
50
Traffic (MB per image)

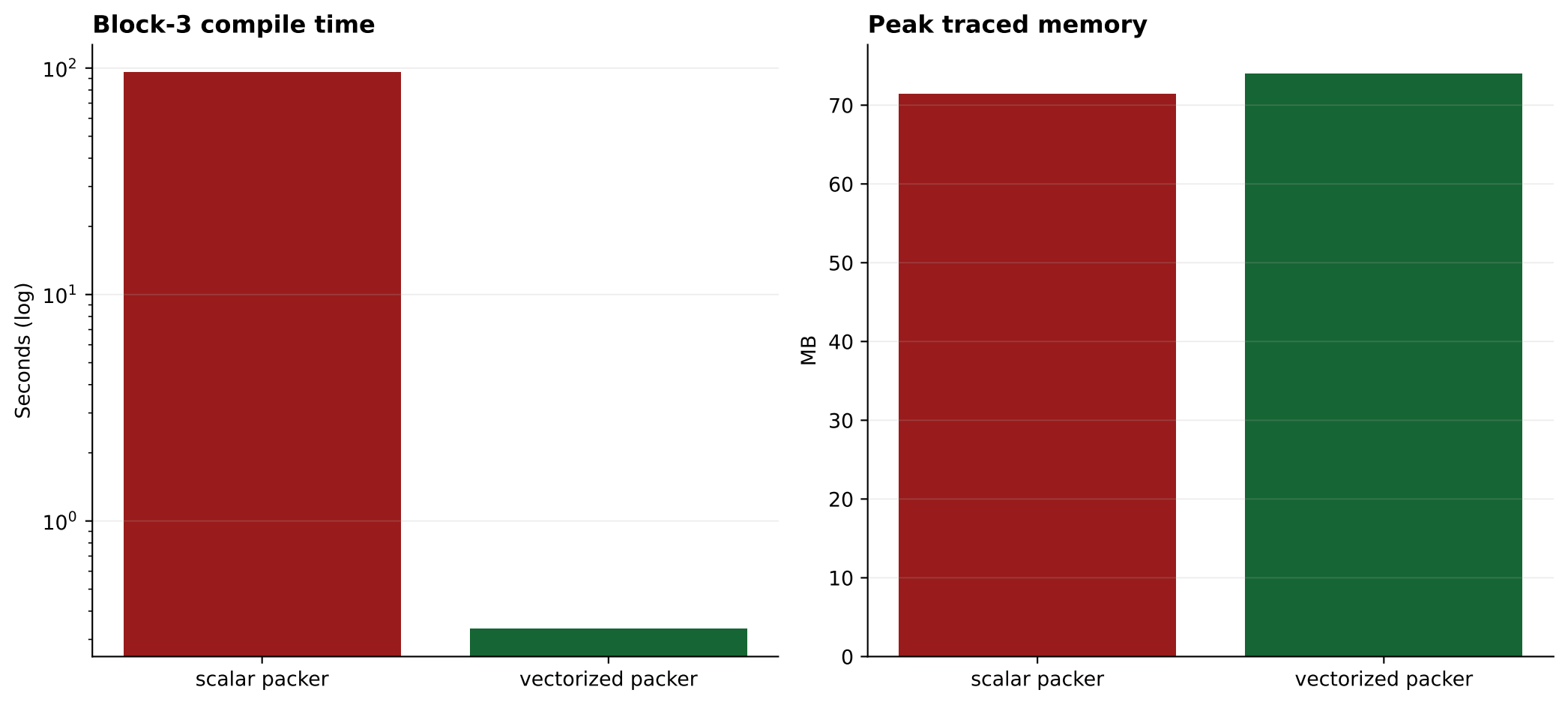

MAC-equivalent share by arithmetic mode
u8_mac
dynamic_u8_mac
1.4%
98.6%
binary_spike_add


Matrix shape landscape
binary_spike_add
dynamic_u8_mac
u8_mac
Output columns N
Reduction K
10^1
10^2
10^3
10^2
10^3


Per-projection INT8 weight scale
Scale
3×10^{-3}
2×10^{-3}
Maximum represented requantization-factor error
Absolute factor error
4×10^{-4}
3×10^{-4}
2×10^{-4}
10^{-4}
6×10^{-5}
4×10^{-5}
3×10^{-5}
2×10^{-5}
patch_embed.block0
patch_embed.block1
patch_embed.block2
patch_embed.block3
patch_embed.block4
block.0.attn.q
block.0.attn.k
block.0.attn.v
block.0.attn.proj
block.0.mlp.fc1
block.0.mlp.fc2
block.1.attn.q
block.1.attn.k
block.1.attn.v
block.1.attn.proj
block.1.mlp.fc1
block.1.mlp.fc2
block.2.attn.q
block.2.attn.k
block.2.attn.v
block.2.attn.proj
block.2.mlp.fc1
block.2.mlp.fc2
block.3.attn.q
block.3.attn.k
block.3.attn.v
block.3.attn.proj
block.3.mlp.fc1
block.3.mlp.fc2


All exported INT8 weights
Count (log)
10^5
10^4
10^3
10^2
10^1
10^0
−100
−50
0
50
100
Integer weight


Twenty most frequent codes
Count
300000
250000
200000
150000
100000
50000
0
9
-10
8
-9
7
-8
6
-7
5
-6
4
-5
-4
3
-3
2
-2
1
-1
0
Integer weight


Block-3 compile time
Seconds (log)
10^2
10^1
10^0
scalar packer
vectorized packer
Peak traced memory
MB
0
10
20
30
40
50
60
70
scalar packer
vectorized packer

Full-Backbone Training Schedule

Target: complete exported integer inference graph

Execution operators covered: 43/43; uncovered 0

Learned affine groups / dynamic attention products: 30 / 8

Trainable deployed parameter elements: 9,295,978

Backward partial tiles / signed multiplications: 14,742,724 / 7,474,847,252

Fabric SGDW update commands: 1,161,998

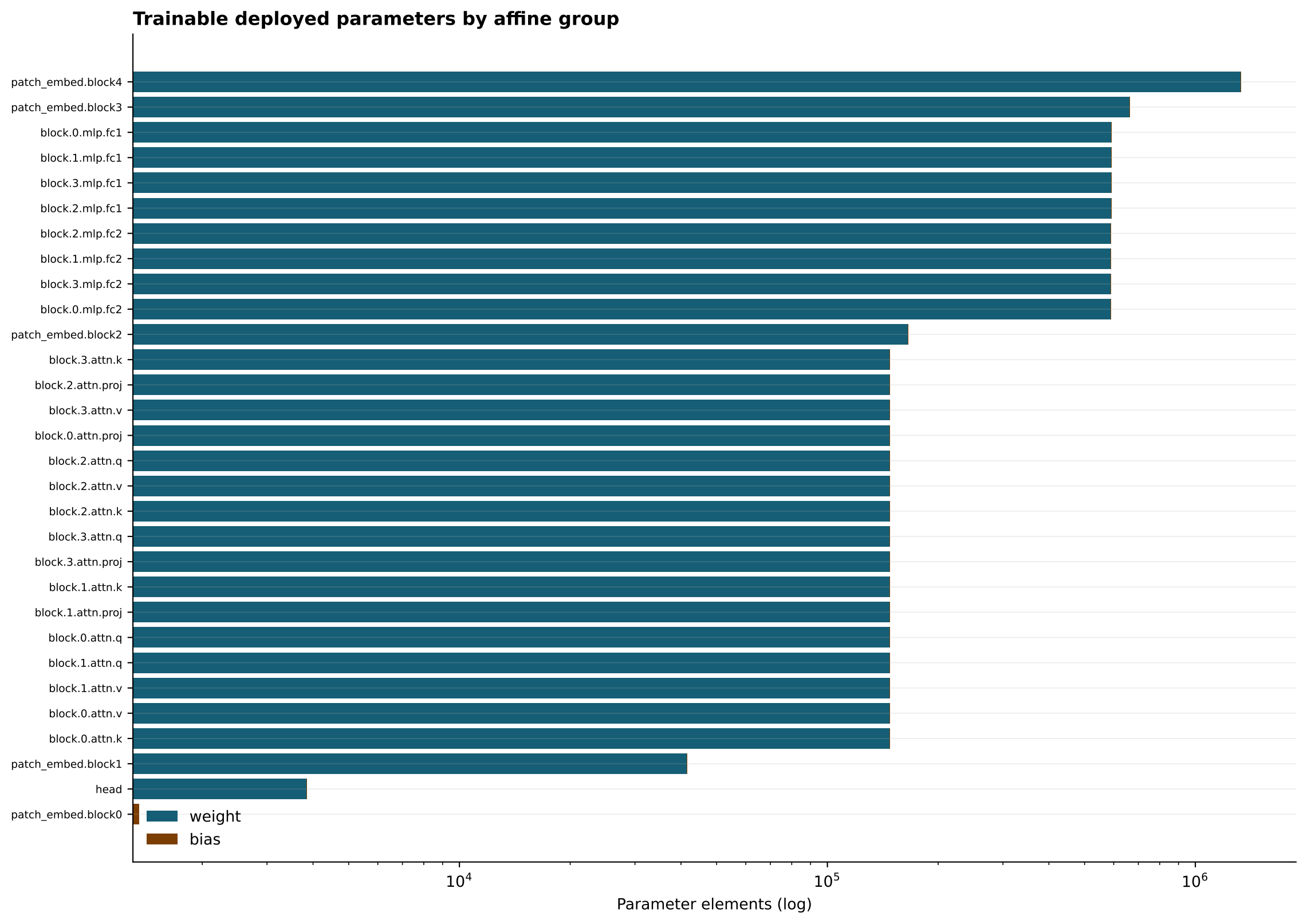
Quantizer derivative: straight-through estimator with fixed exported scales

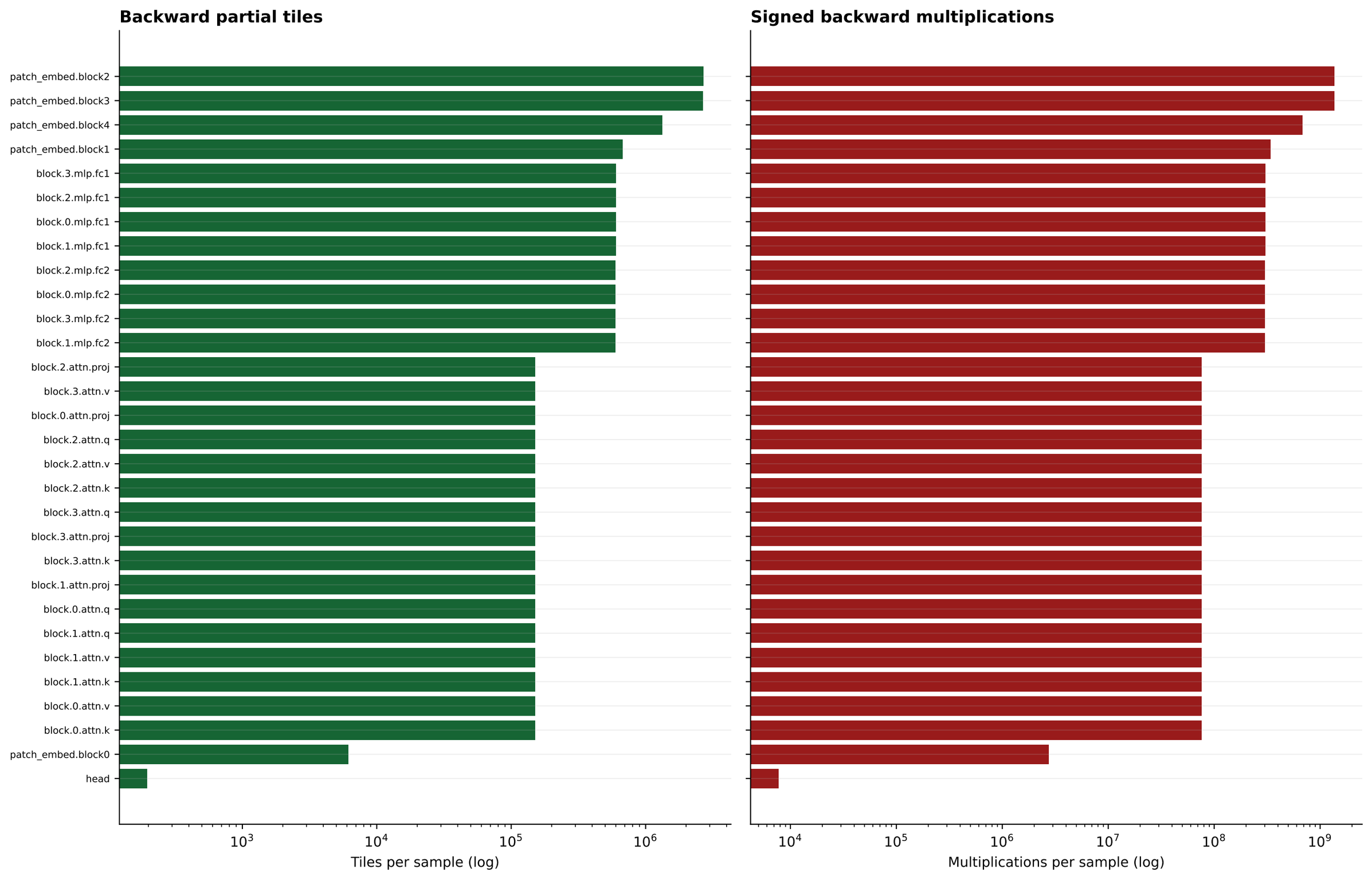
LIF derivative: SpikingJelly sigmoid surrogate alpha=4, hard reset, detach_reset=True

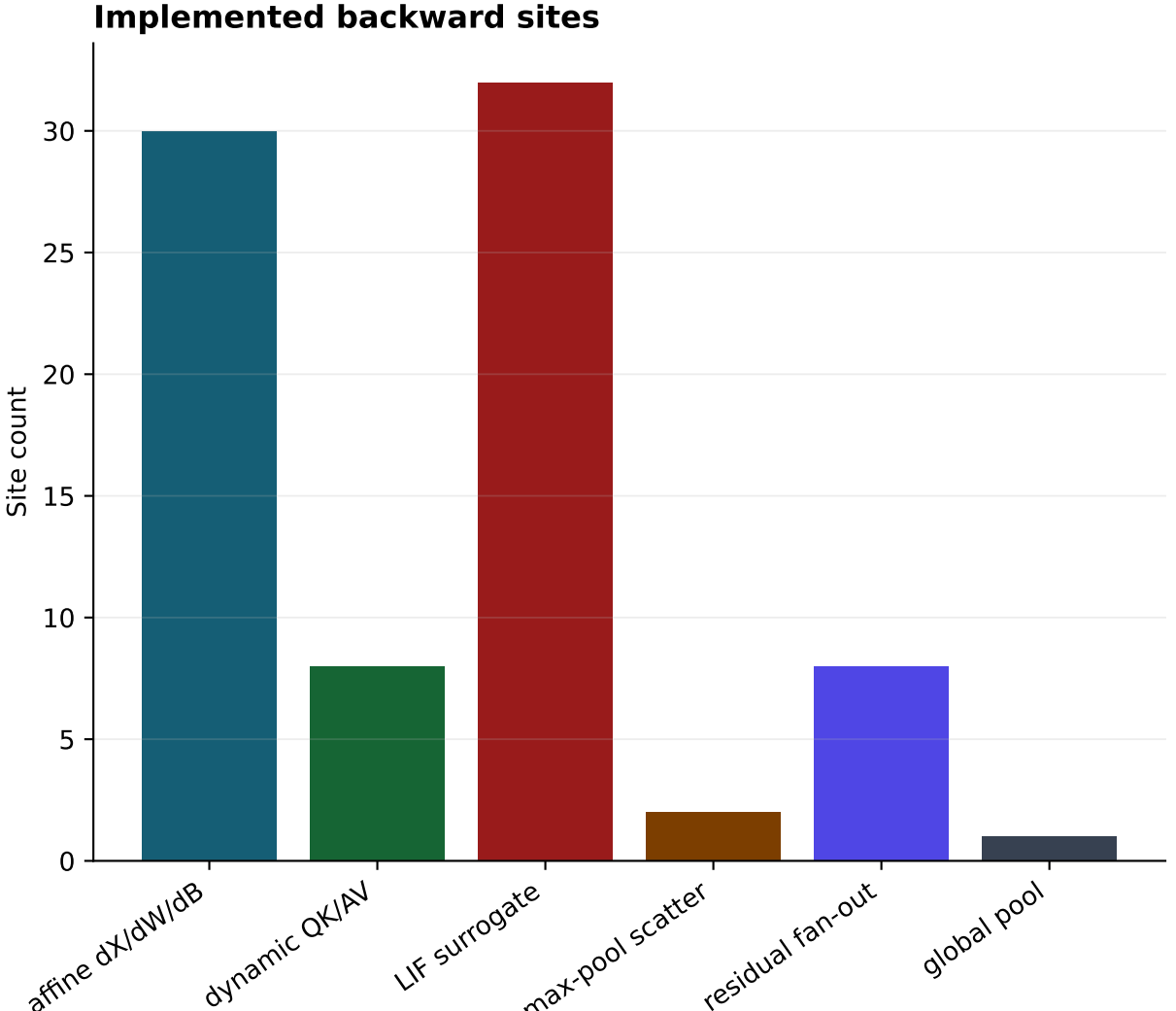
Temporal schedule: four-step BPTT in reverse timestep order through cached membrane traces

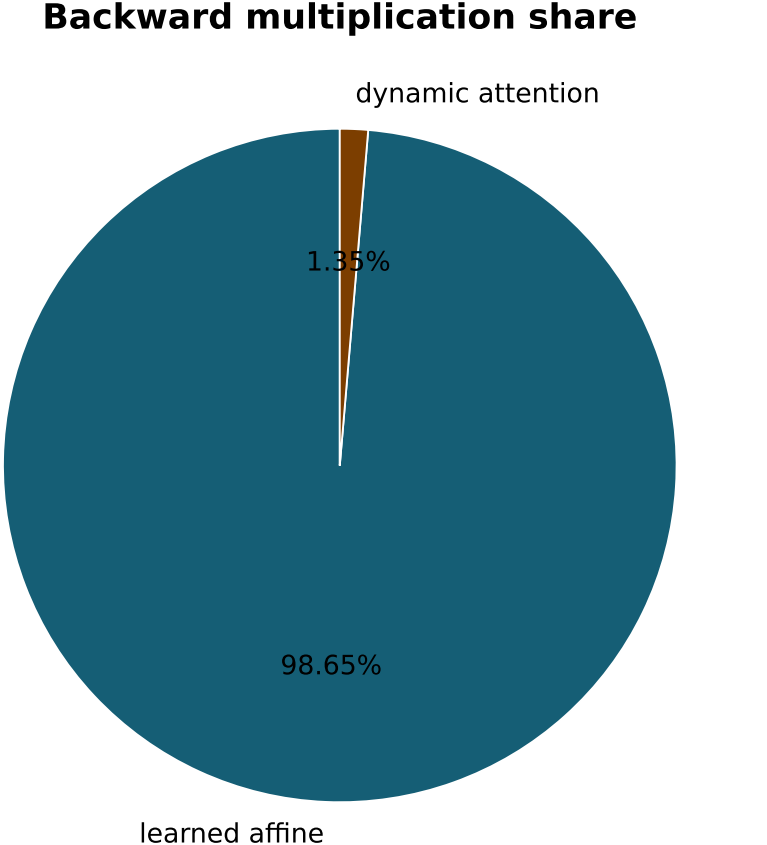
Optimizer: signed fixed-point SGDW with momentum; weight decay is applied to the update direction and excluded from retained velocity

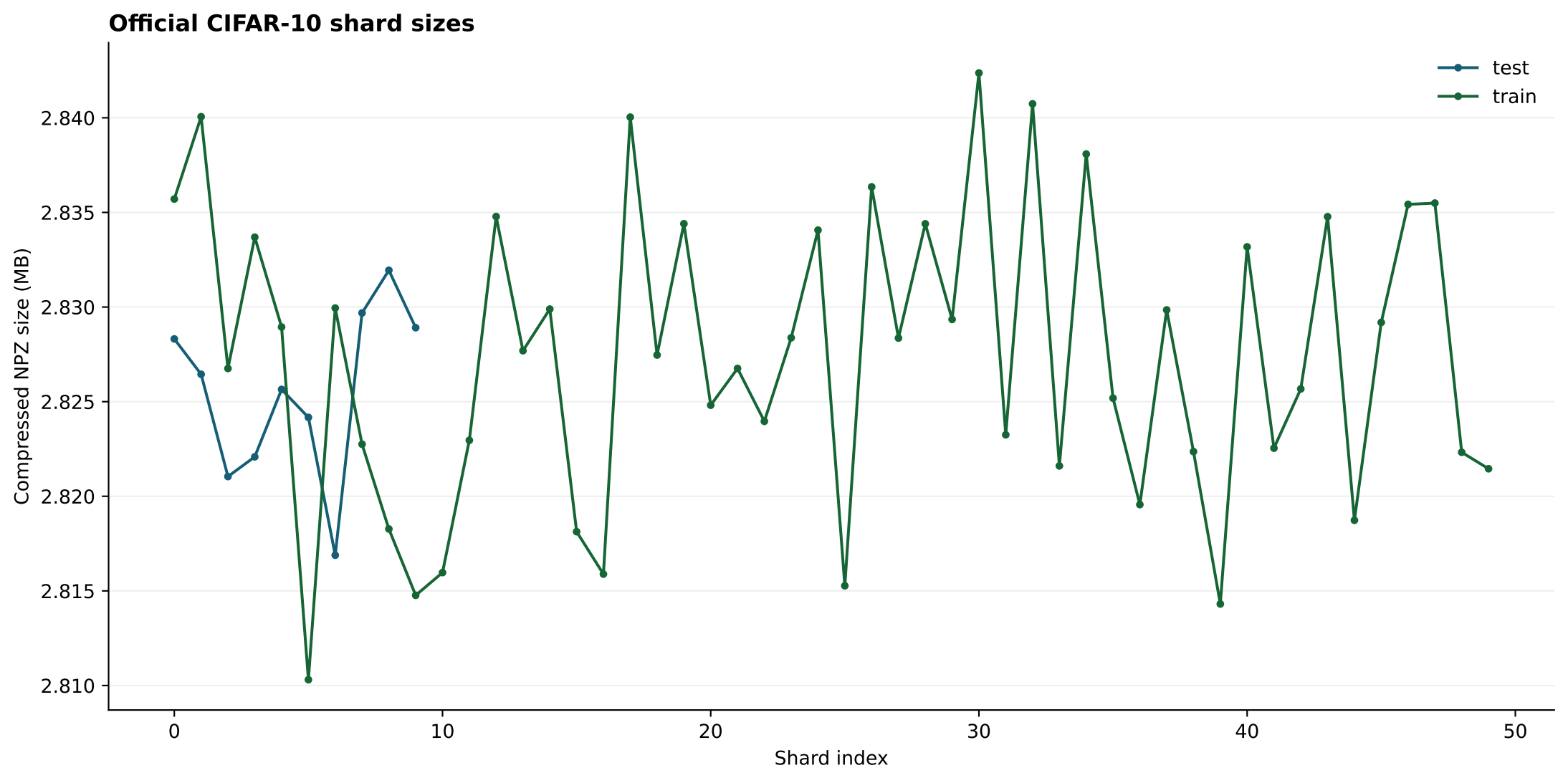
Parameter residency: PYNQ DDR master tensors; arithmetic and updates in programmable logic


Trainable deployed parameters by affine group
patch_embed.block4
patch_embed.block3
block.0.mlp.fc1
block.1.mlp.fc1
block.3.mlp.fc1
block.2.mlp.fc1
block.2.mlp.fc2
block.1.mlp.fc2
block.3.mlp.fc2
block.0.mlp.fc2
patch_embed.block2
block.3.attn.k
block.2.attn.proj
block.3.attn.v
block.0.attn.proj
block.2.attn.q
block.2.attn.v
block.2.attn.k
block.3.attn.q
block.3.attn.proj
block.1.attn.k
block.1.attn.proj
block.0.attn.q
block.1.attn.q
block.1.attn.v
block.0.attn.v
block.0.attn.k
patch_embed.block1
head
patch_embed.block0
weight
bias
10^4
10^5
10^6
Parameter elements (log)

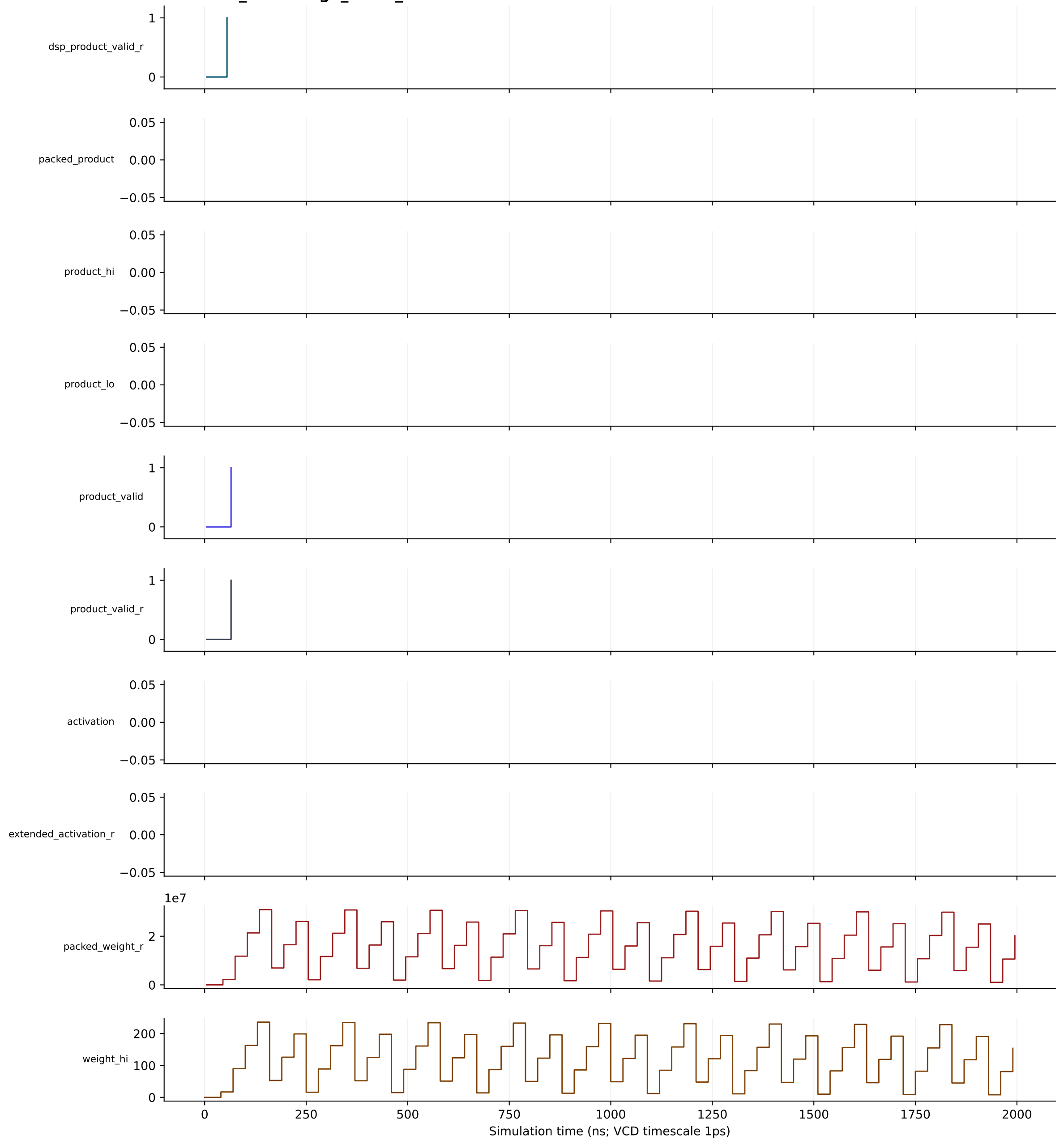

Backward partial tiles
Signed backward multiplications
patch_embed.block2
patch_embed.block3
patch_embed.block4
patch_embed.block1
block.3.mlp.fc1
block.2.mlp.fc1
block.0.mlp.fc1
block.1.mlp.fc1
block.2.mlp.fc2
block.0.mlp.fc2
block.3.mlp.fc2
block.1.mlp.fc2
block.2.attn.proj
block.3.attn.v
block.0.attn.proj
block.2.attn.q
block.2.attn.v
block.2.attn.k
block.3.attn.q
block.3.attn.proj
block.3.attn.k
block.1.attn.proj
block.0.attn.q
block.1.attn.q
block.1.attn.v
block.1.attn.k
block.0.attn.v
block.0.attn.k
patch_embed.block0
head
10^3
10^4
10^5
10^6
Tiles per sample (log)
10^4
10^5
10^6
10^7
10^8
10^9
Multiplications per sample (log)


Implemented backward sites
Site count
0
5
10
15
20
25
30
affine dX/dW/dB
dynamic QK/AV
LIF surrogate
max-pool scatter
residual fan-out
global pool

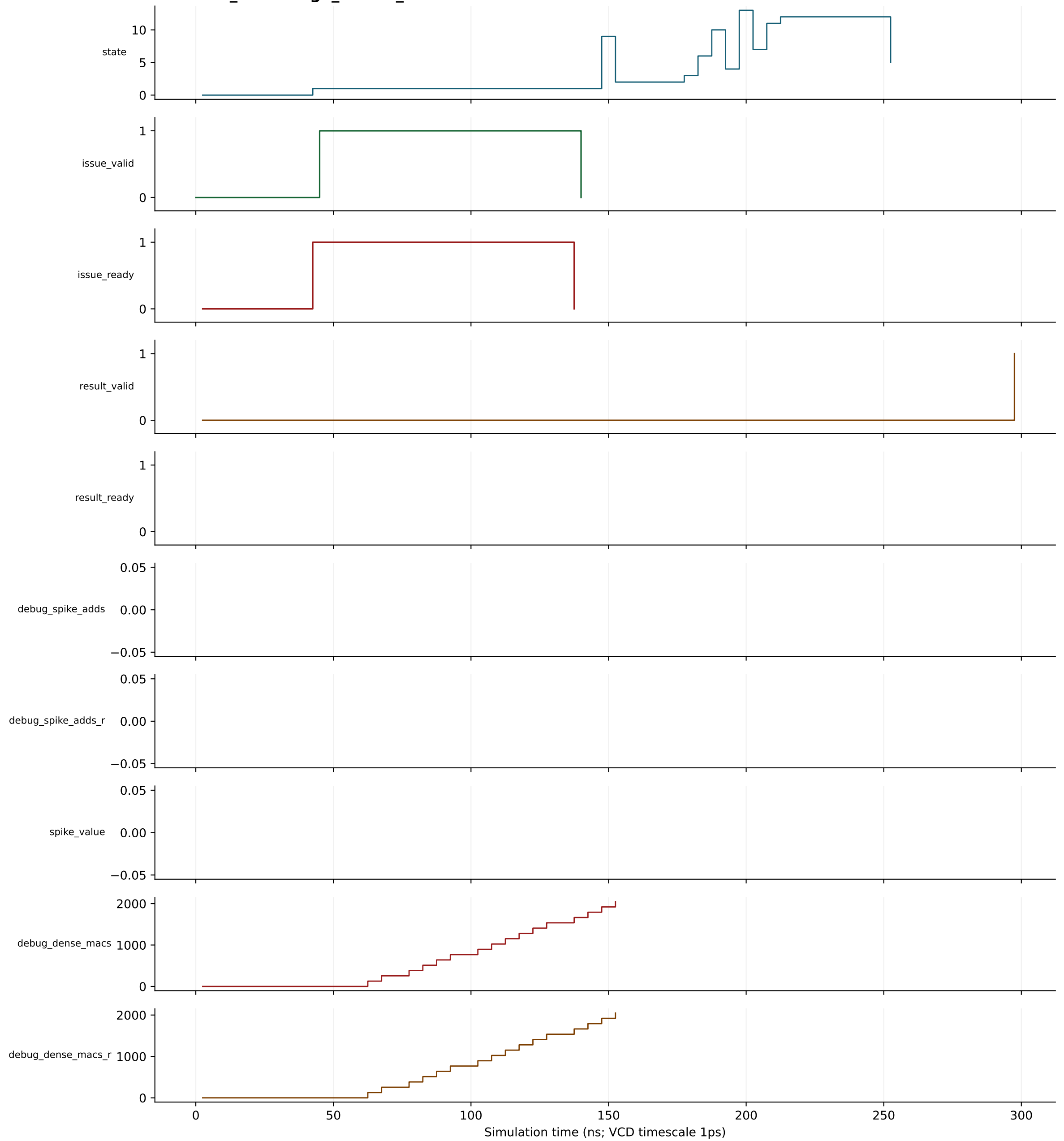

Backward multiplication share
dynamic attention
1.35%
98.65%
learned affine


Official CIFAR-10 shard sizes
test
train
Compressed NPZ size (MB)
2.840
2.835
2.830
2.825
2.820
2.815
2.810
0
10
20
30
40
50
Shard index

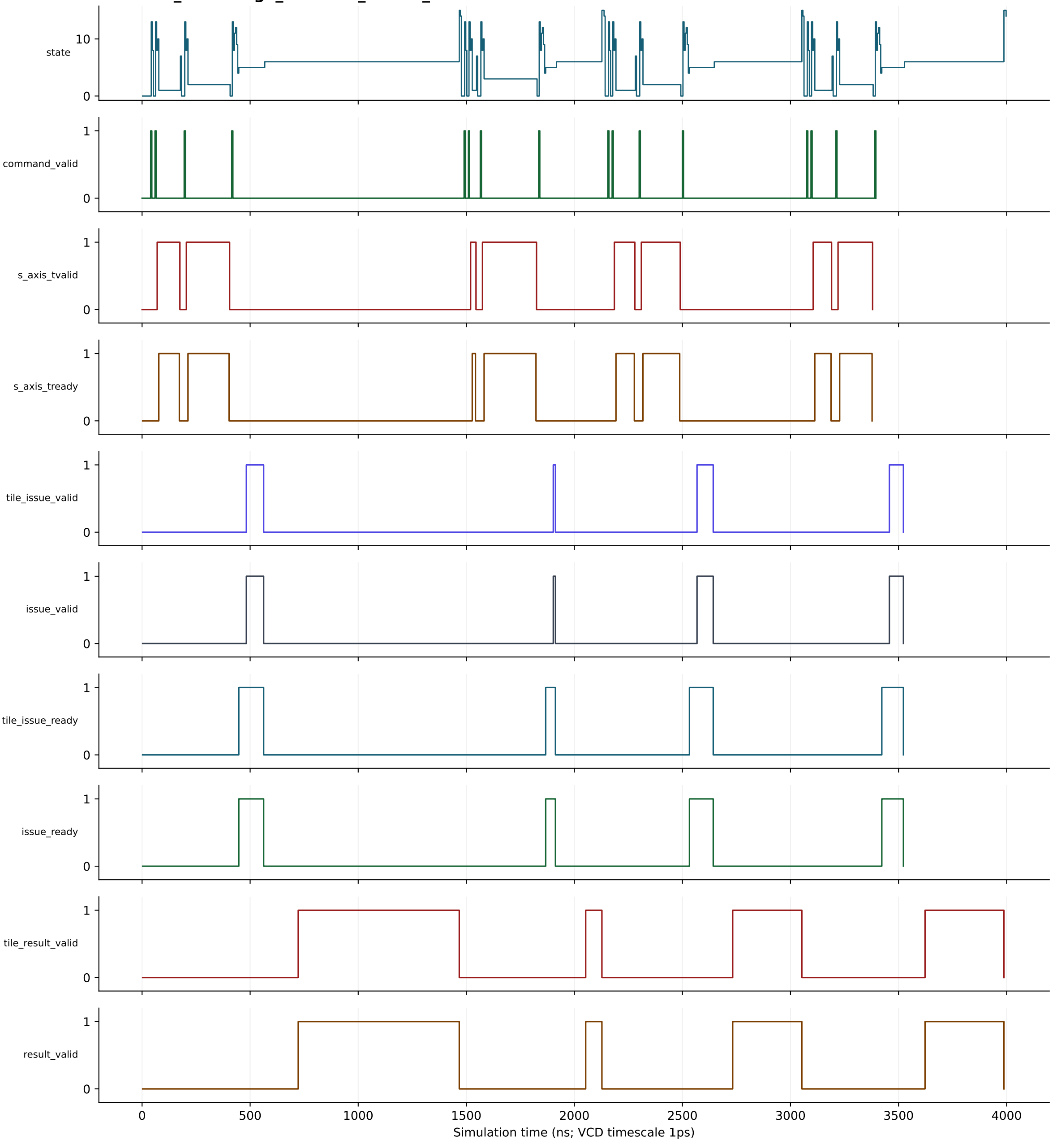
XSim: tb_neuraedge_mac2_u8



XSim: tb_neuraedge_tensor_tile



XSim: tb_neuraedge_buffered_tensor_core



XSim: tb_neuraedge_alpha_xnor

state

output_valid

output_ready

final_spike_count_r

key_spikes

query_spikes

score_spike_operand_r

spike_chunk_count_r

spike_count_r

spike_mask_r

Simulation time (ns; VCD timescale 1ps)

XSim: tb_neuraedge_vector_alu

state

output_valid

output_ready

OP_BINARY_SPIKE

OP_TERNARY_SPIKE

multiplier_product_hh

multiplier_product_hl

multiplier_product_lh

multiplier_product_ll

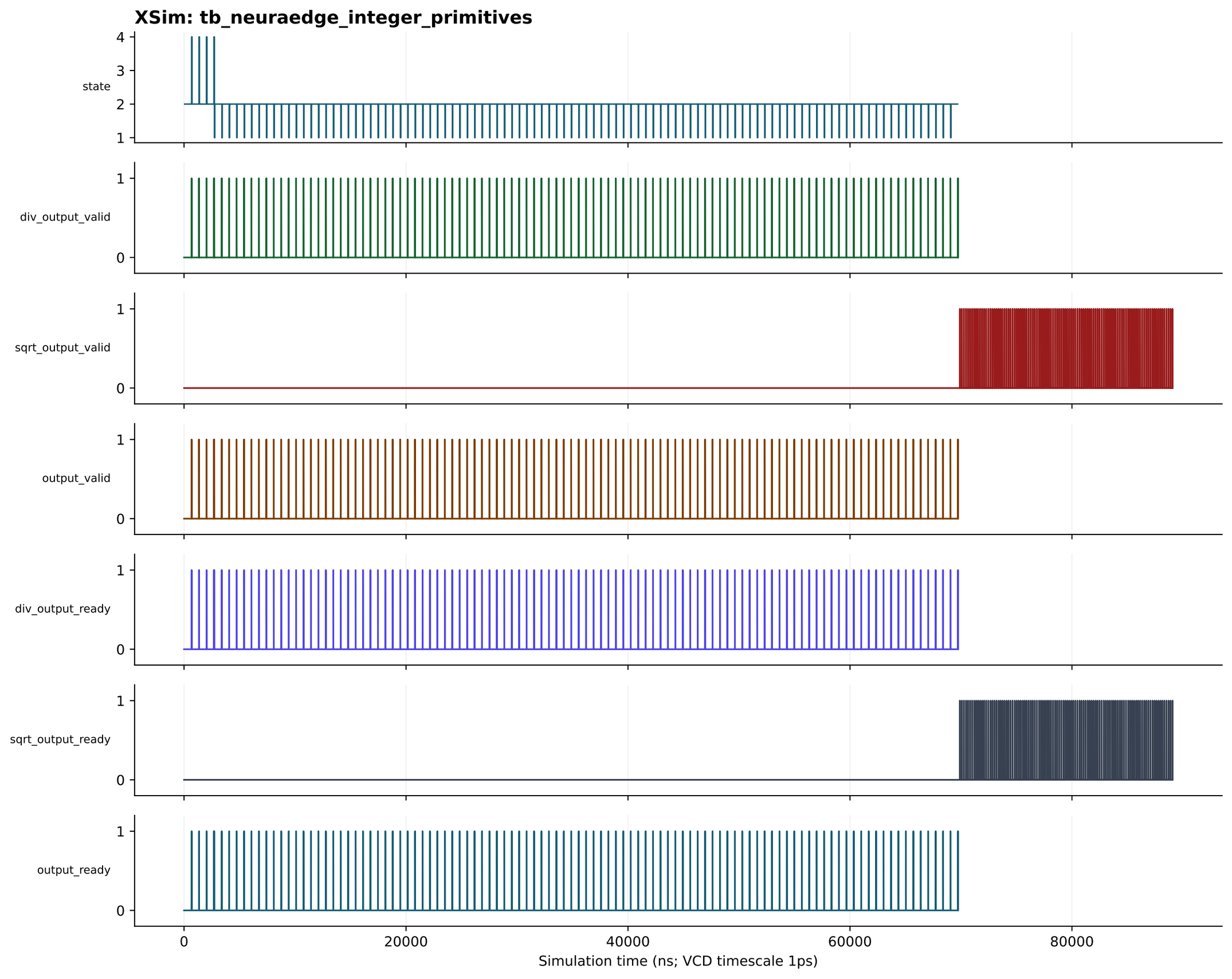
product_cross_r

1e9

1e10

0 25000 50000 75000 100000 125000 150000 175000

Simulation time (ns; VCD timescale 1ps)


XSim: tb_neuraedge_integer_primitives
state
4
3
2
1
div_output_valid
1
0
sqrt_output_valid
1
0
output_valid
1
0
div_output_ready
1
0
sqrt_output_ready
1
0
output_ready
1
0
0
20000
40000
60000
80000
Simulation time (ns; VCD timescale 1ps)

XSim: tb_neuraedge_normalization_units

state

ln_result_valid

result_valid

ln_result_ready

result_ready

divider_output_valid

sqrt_output_valid

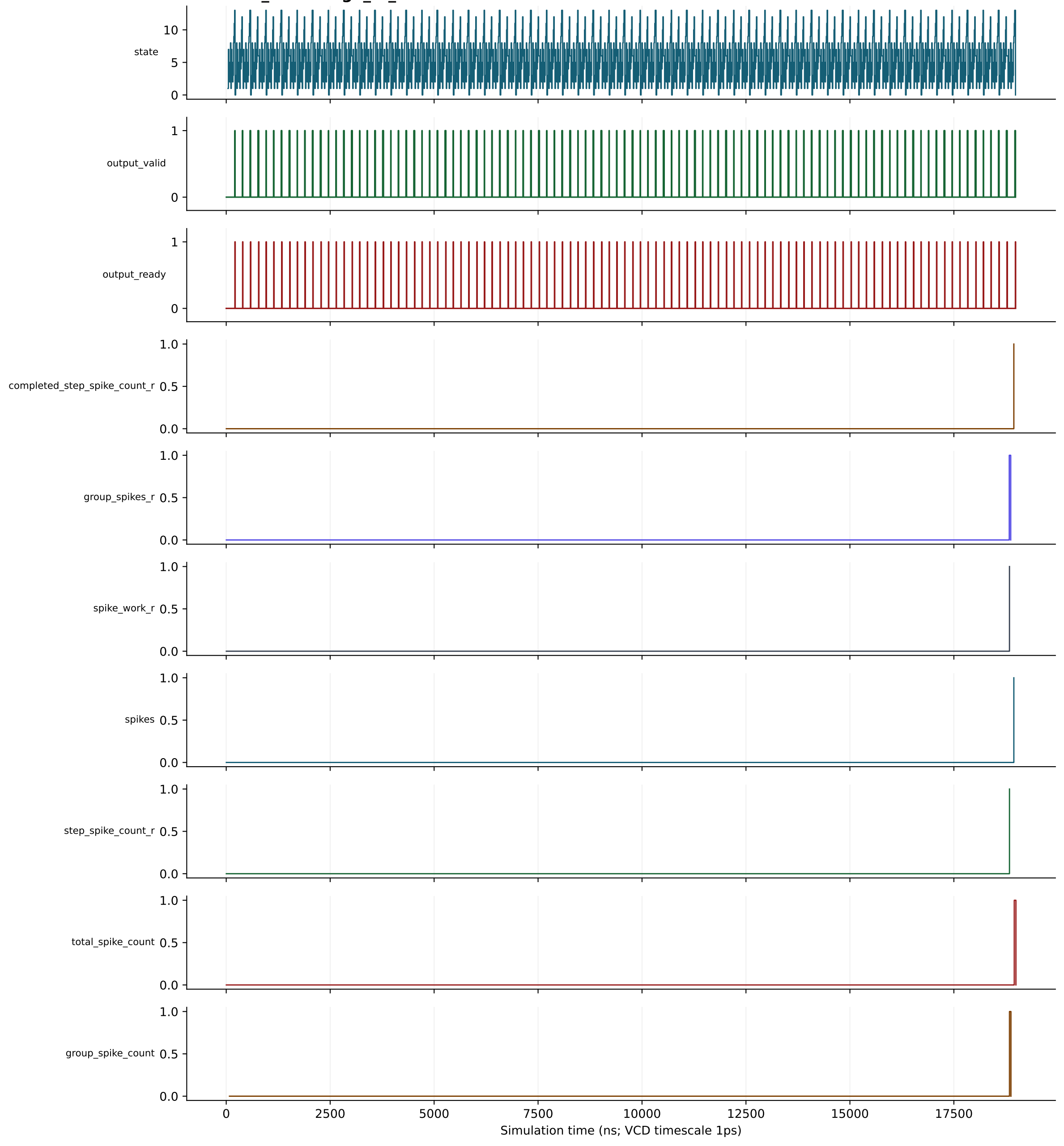
output_valid

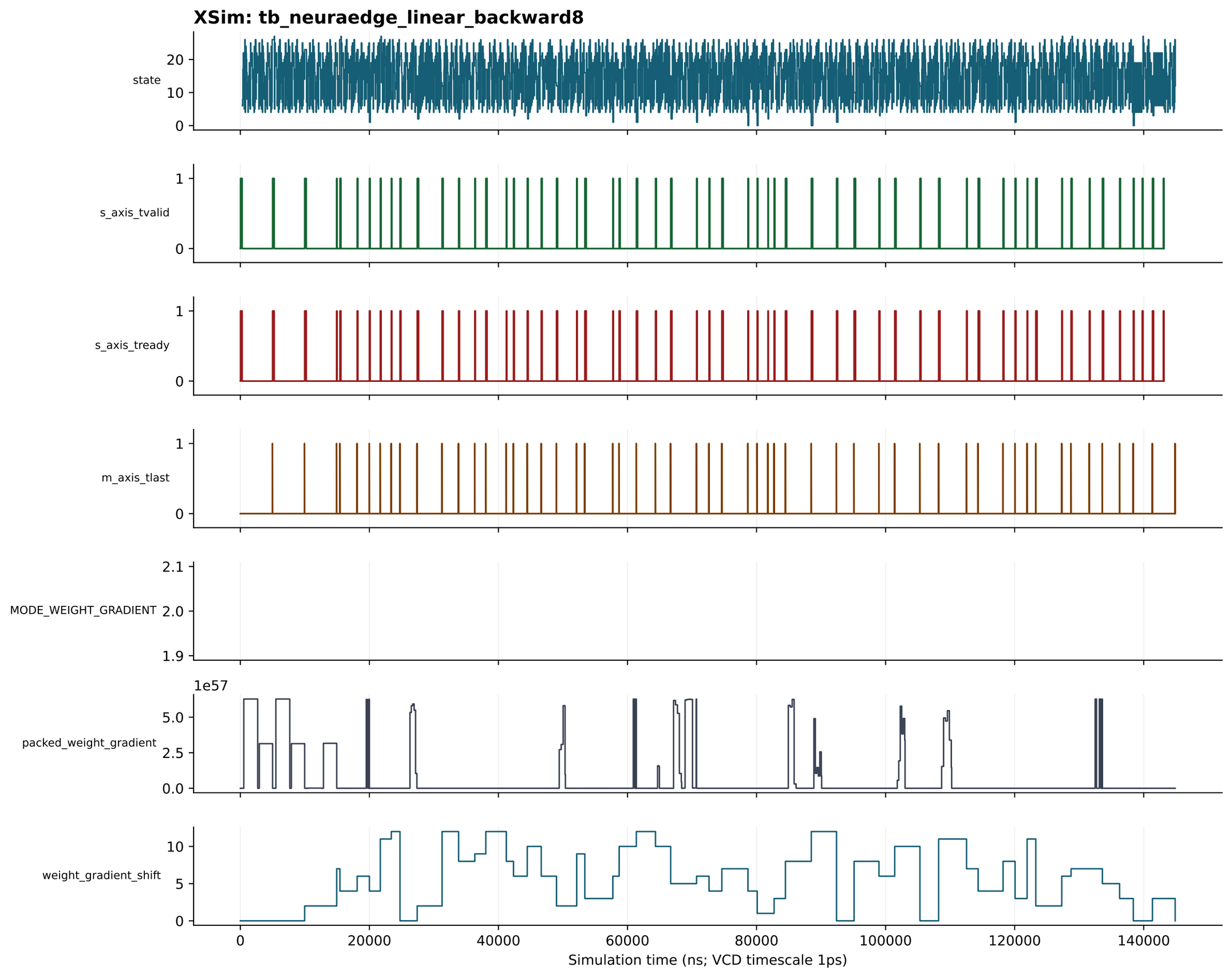
divider_output_ready

sqrt_output_ready

Simulation time (ns; VCD timescale 1ps)

XSim: tb_neuraedge_lif_bank




XSim: tb_neuraedge_linear_backward8
state
20
10
0
s_axis_tvalid
1
0
s_axis_tready
1
0
m_axis_tlast
1
0
MODE_WEIGHT_GRADIENT
2.1
2.0
1.9
1e57
packed_weight_gradient
5.0
2.5
0.0
weight_gradient_shift
10
5
0
0
20000
40000
60000
80000
100000
120000
140000
Simulation time (ns; VCD timescale 1ps)

XSim: tb_neuraedge_sgd_optimizer8

state
20
0

output_valid
1
0

output_ready
1
0

update_count
500
250
0

1e38
updated_weight_vector
2
0

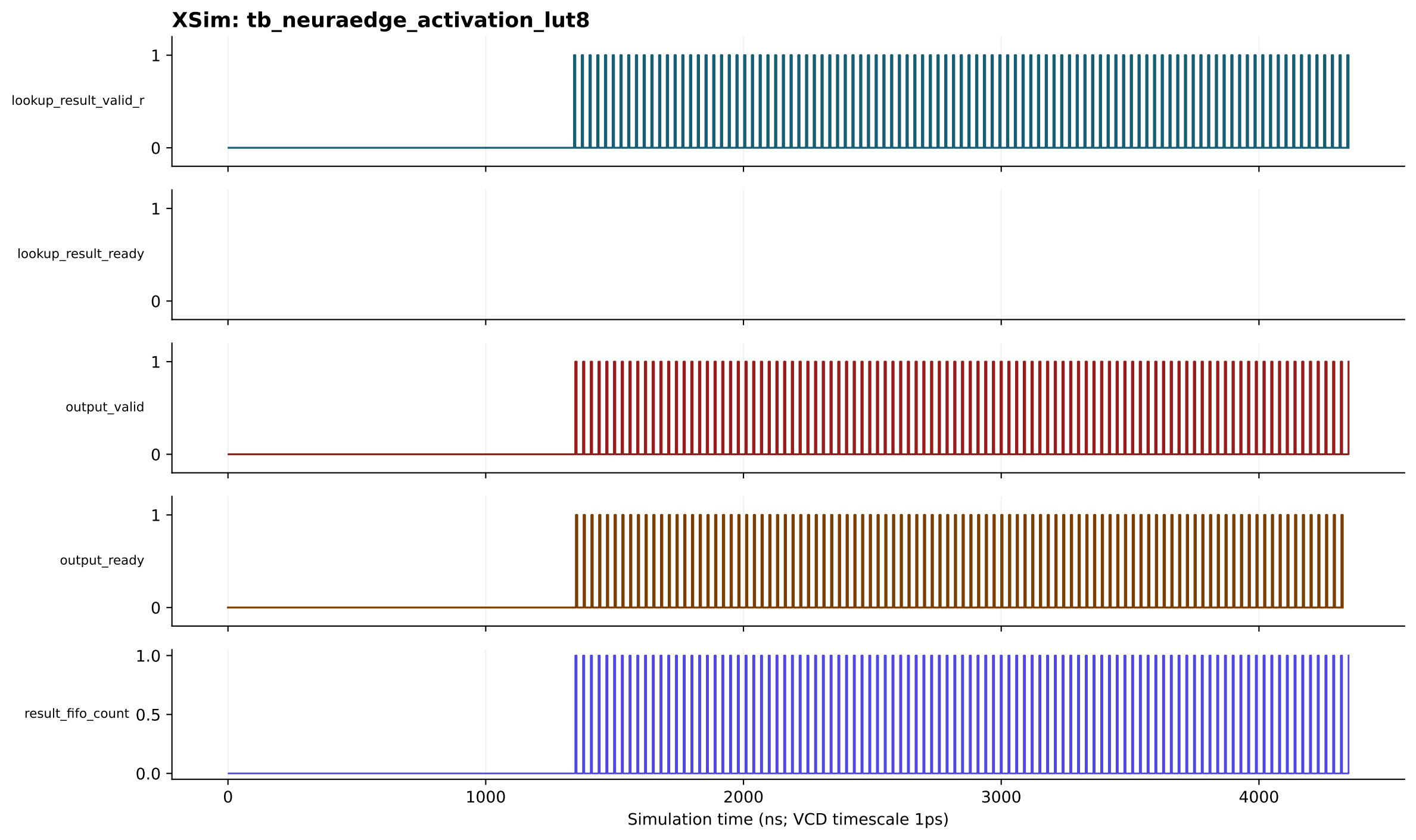
weight_decay_q16
50000
25000
0

weight_decay_r
50000
25000
0

1e38
weight_vector
2
0

1e38
held_weight
2
0

0
2500
5000
7500
10000
12500
15000
17500
20000
Simulation time (ns; VCD timescale 1ps)


XSim: tb_neuraedge_activation_lut8
lookup_result_valid_r
1
0
lookup_result_ready
1
0
output_valid
1
0
output_ready
1
0
result_fifo_count
1.0
0.5
0.0
0
1000
2000
3000
4000
Simulation time (ns; VCD timescale 1ps)

XSim: tb_neuraedge_universal_core

state
20
0

command_valid
1
0

tensor_command_valid
1
0

s_axis_tvalid
1
0

s_axis_tready
1
0

tile_issue_valid
1
0

issue_valid
1
0

tile_issue_ready
1
0

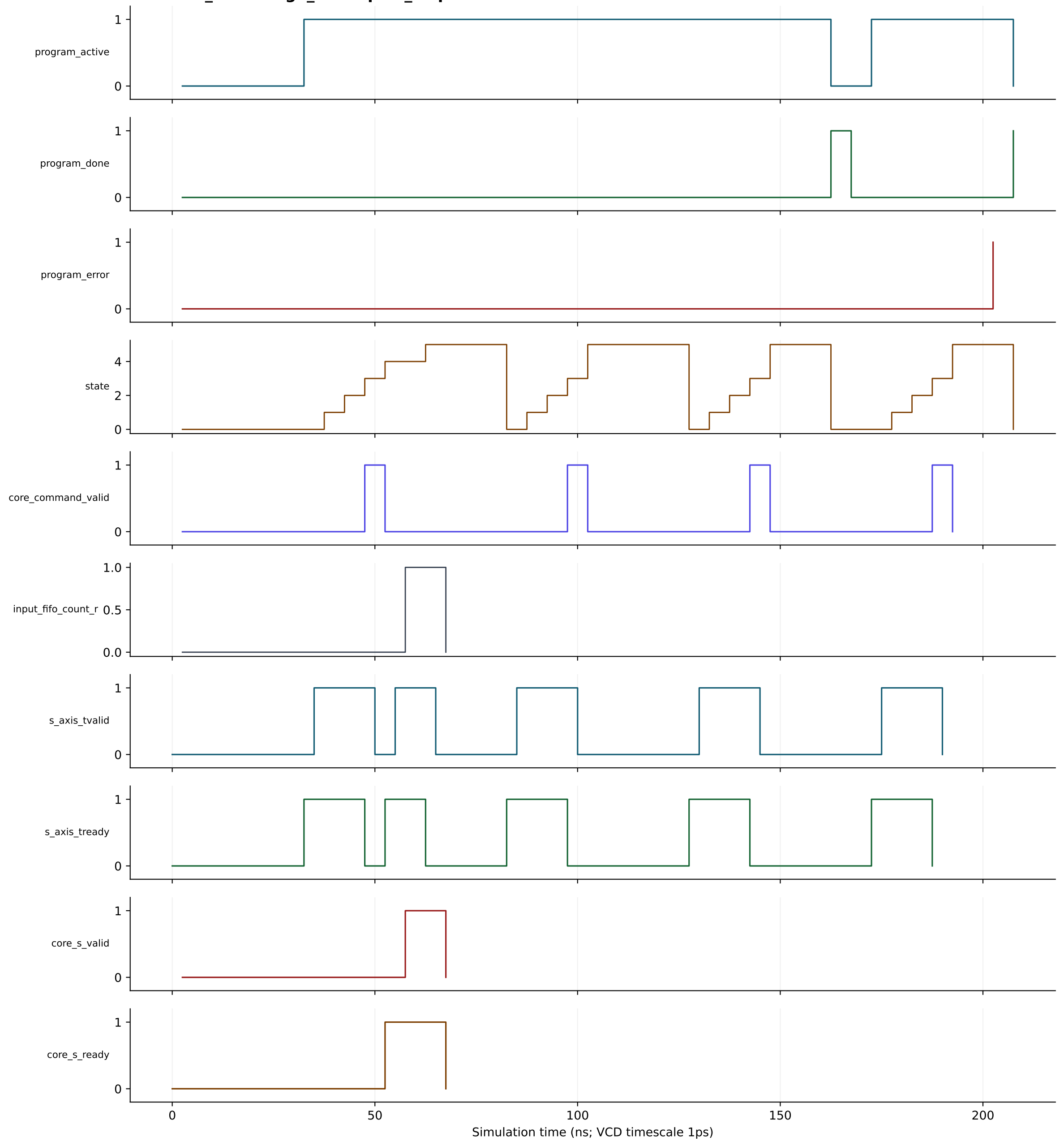
issue_ready
1
0

ln_result_valid
1
0

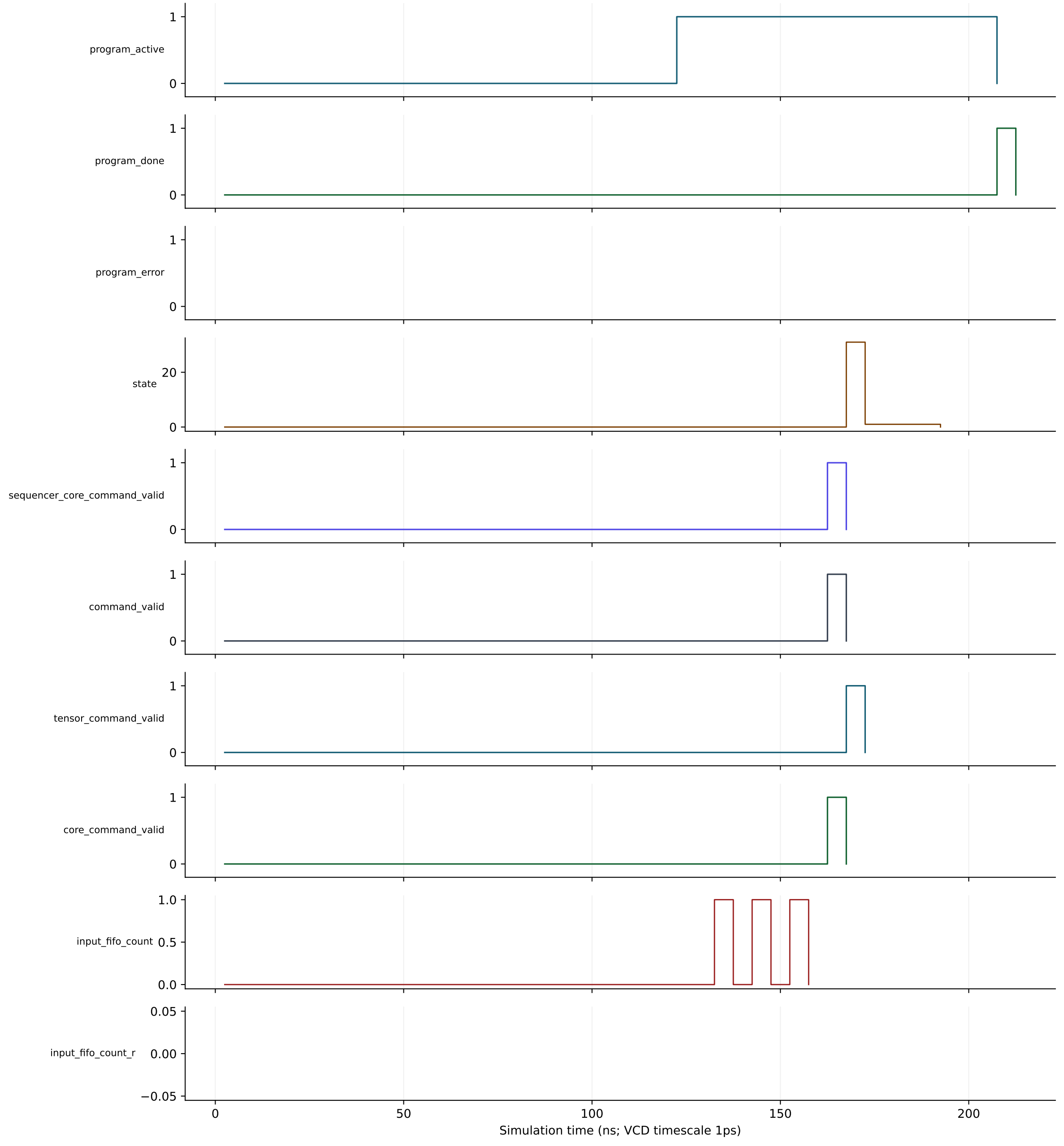
0
2000
4000
6000
8000

Simulation time (ns; VCD timescale 1ps)

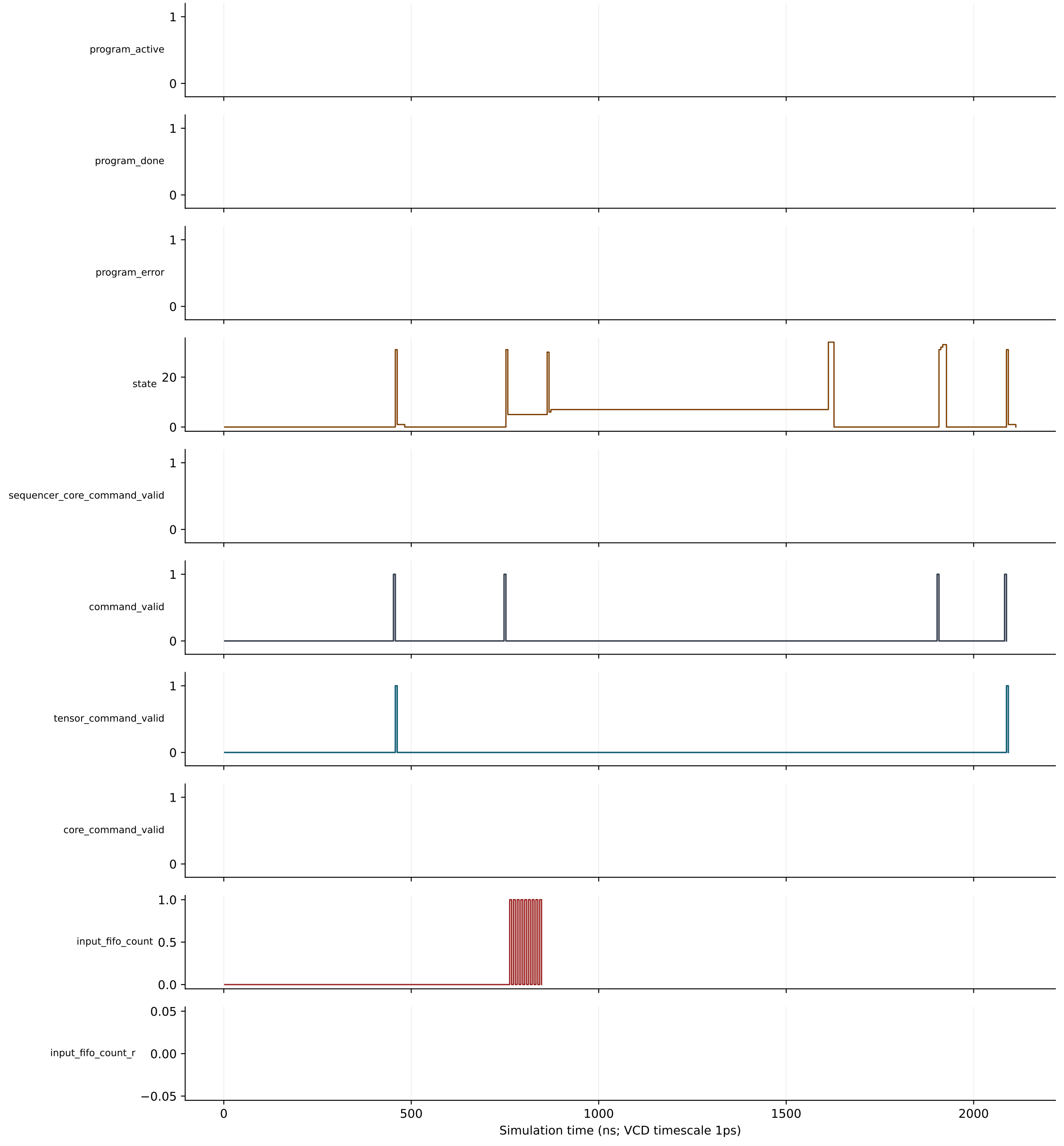
XSim: tb_neuraedge_descriptor_sequencer

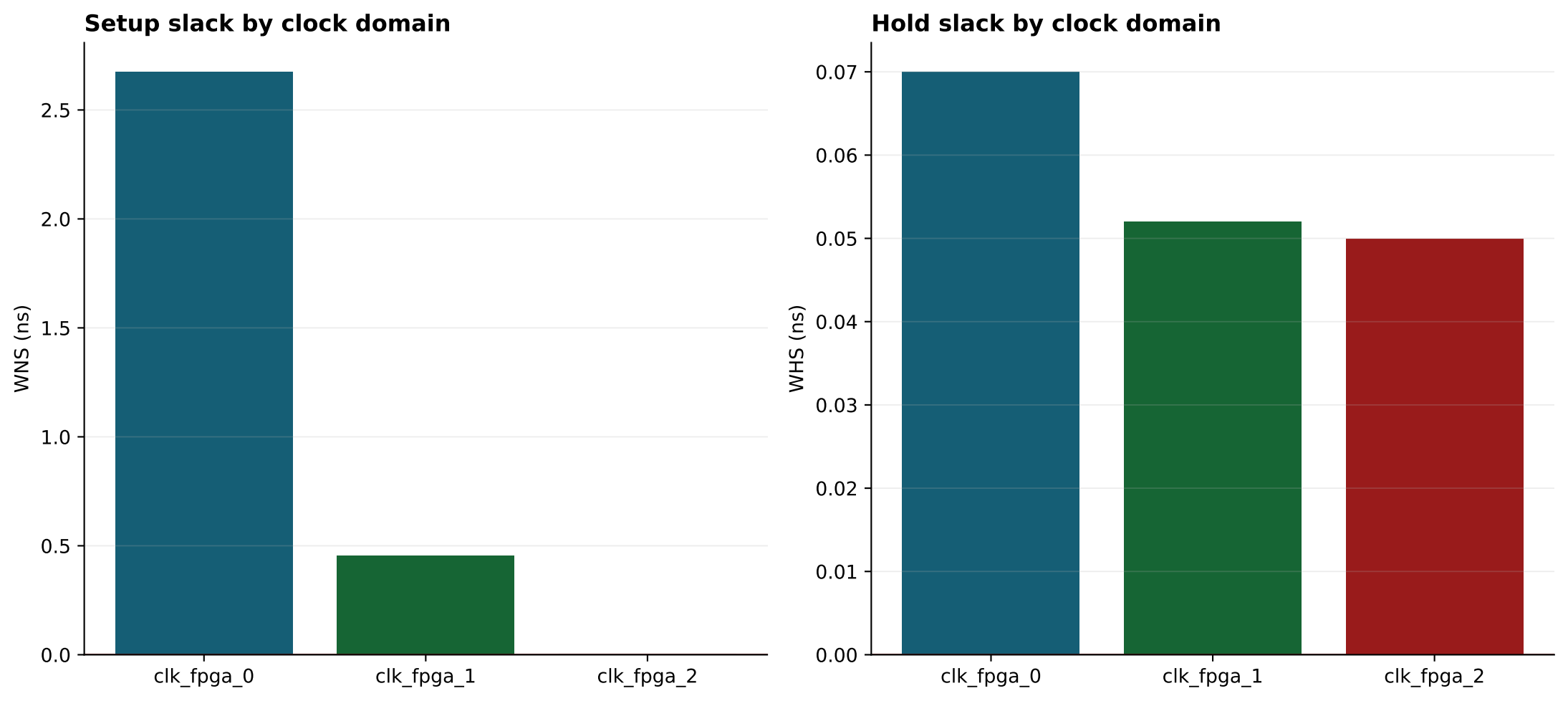


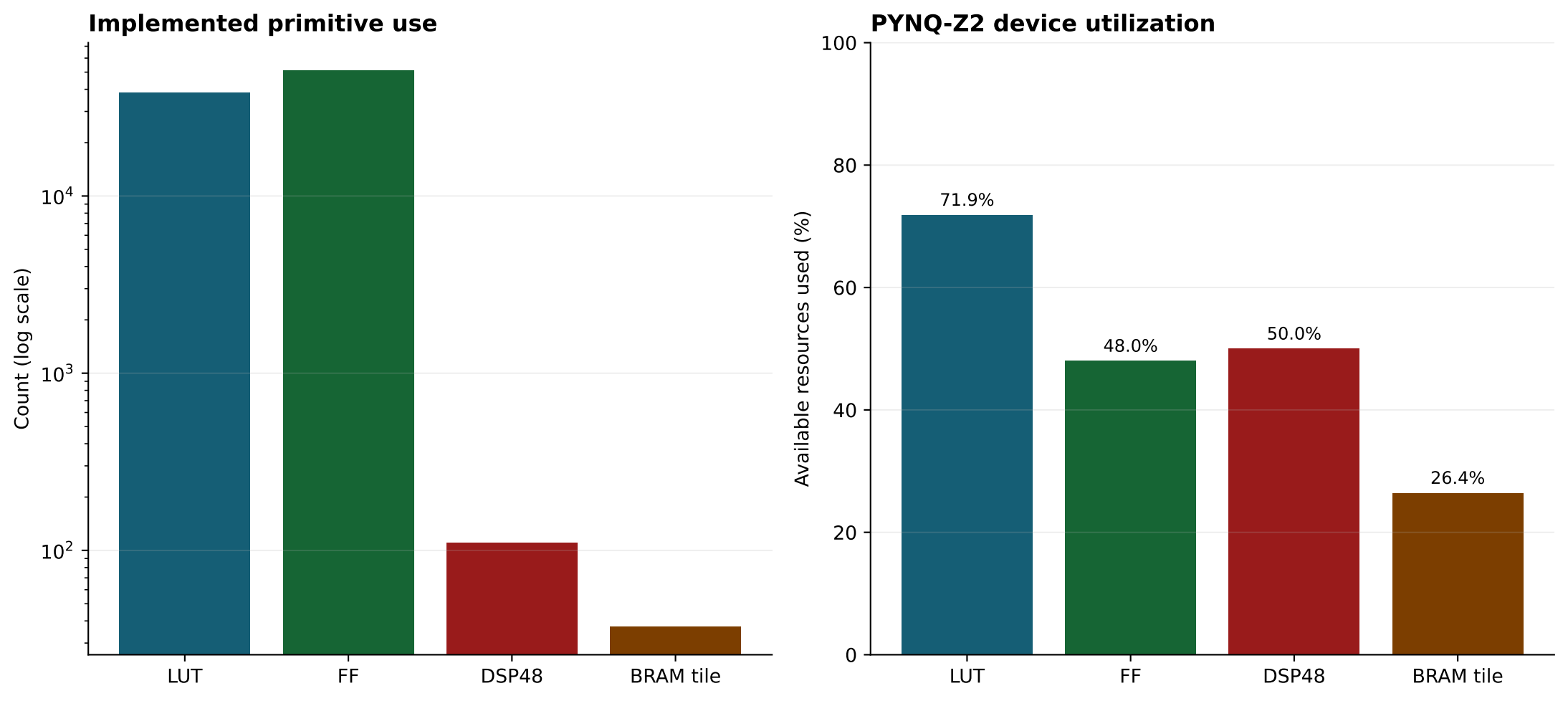
XSim: tb_neuraedge_pynq_descriptor

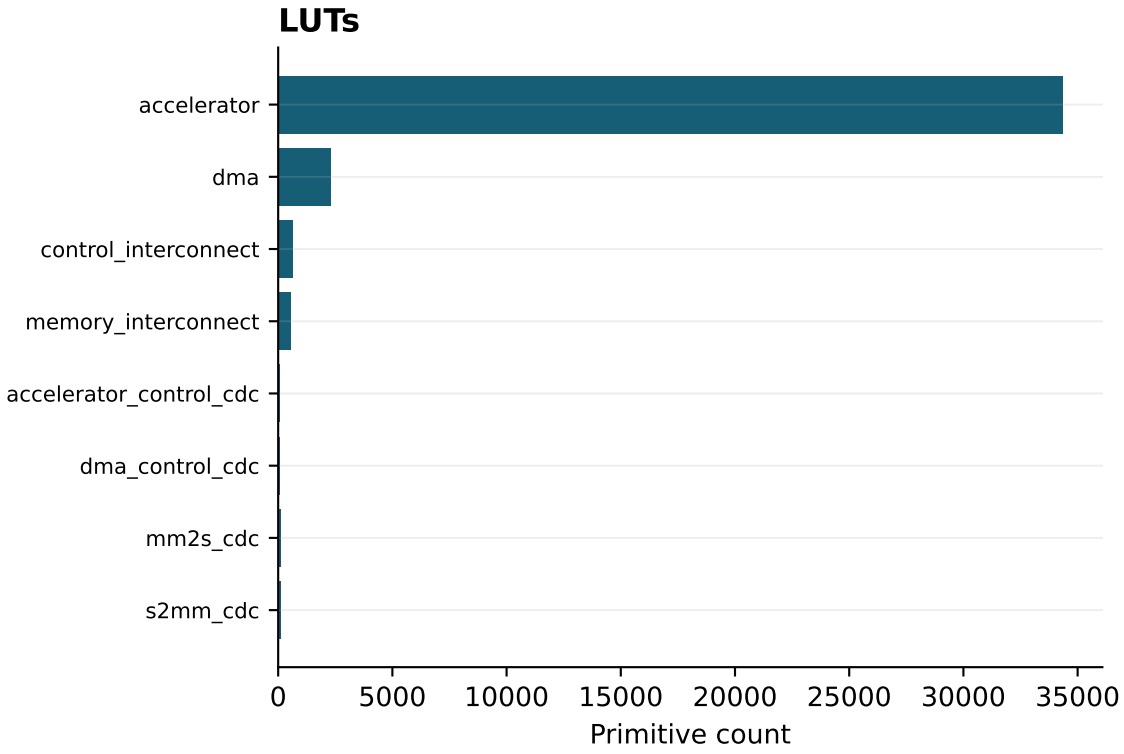


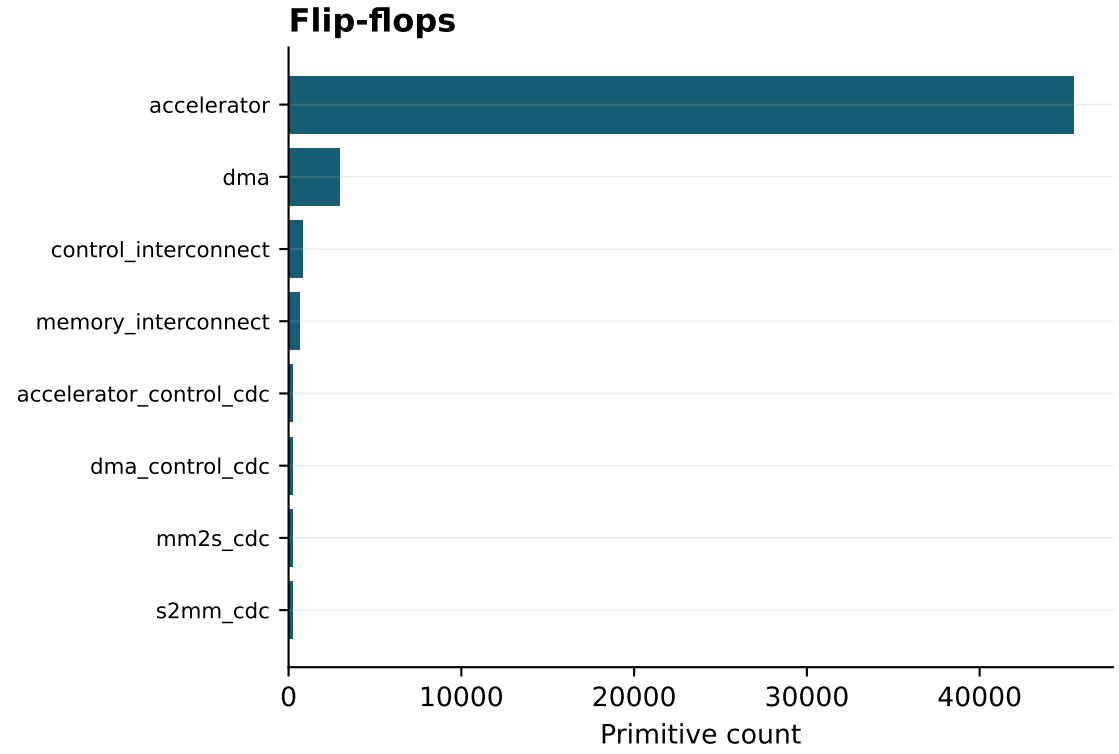
XSim: tb_neuraedge_pynq_axi_top

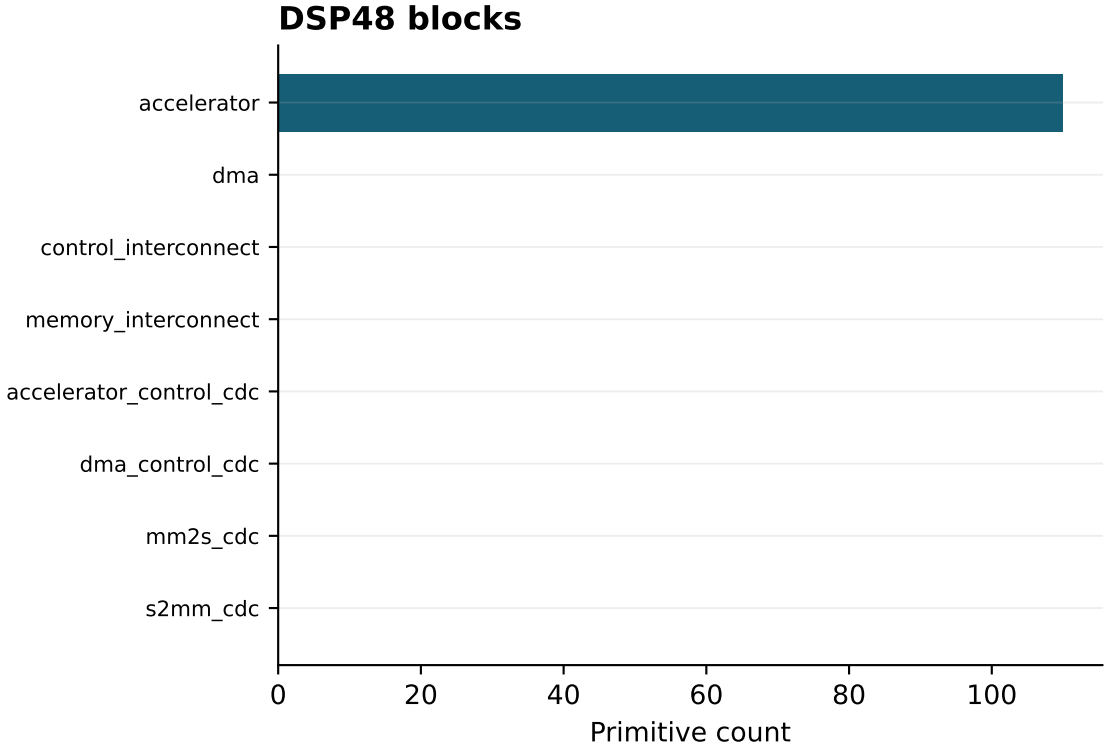


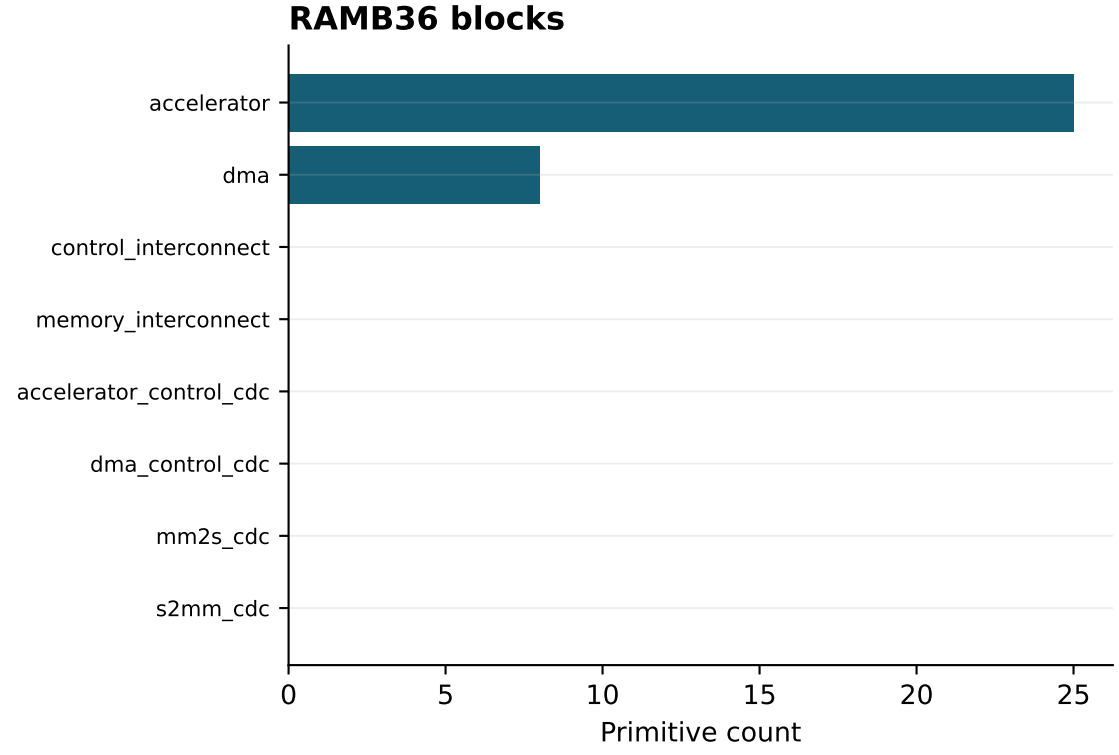

Setup slack by clock domain
WNS (ns)
2.5
2.0
1.5
1.0
0.5
0.0
clk_fpga_0
clk_fpga_1
clk_fpga_2
Hold slack by clock domain
WHS (ns)
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0.00
clk_fpga_0
clk_fpga_1
clk_fpga_2

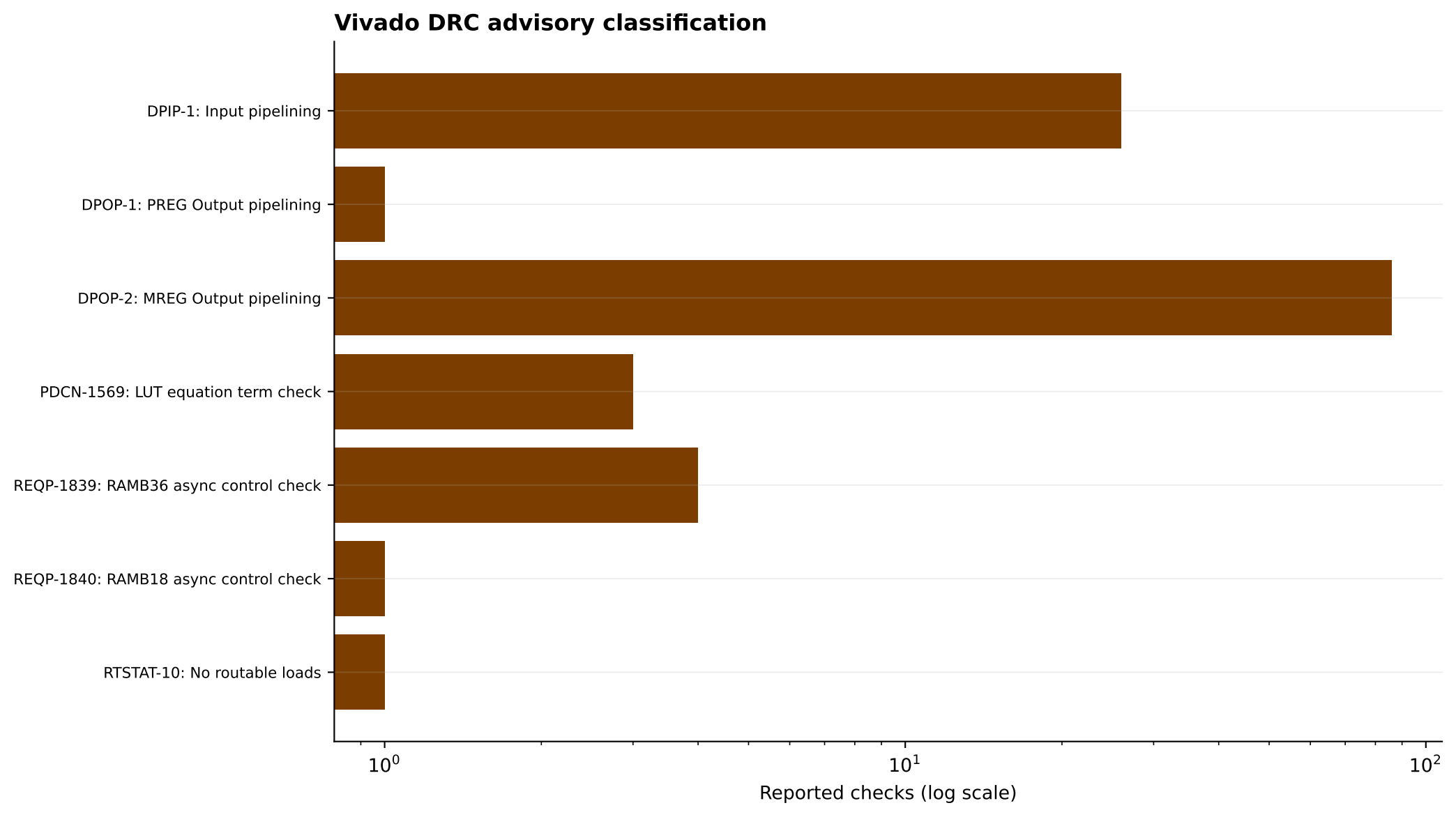

Implemented primitive use
Count (log scale)
10^4
10^3
10^2
LUT
FF
DSP48
BRAM tile
PYNQ-Z2 device utilization
Available resources used (%)
100
80
60
40
20
0
71.9%
48.0%
50.0%
26.4%
LUT
FF
DSP48
BRAM tile

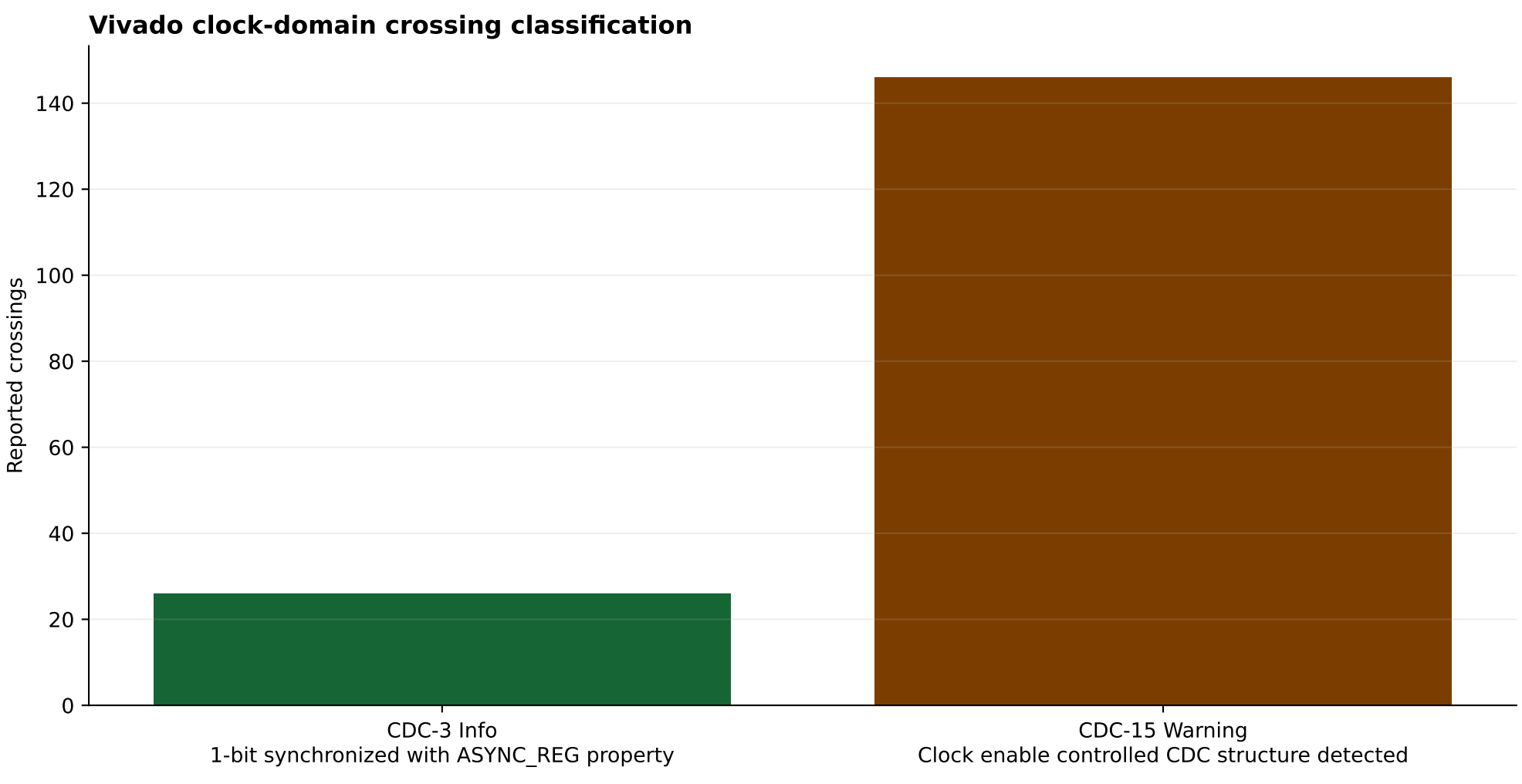

LUTs
accelerator
dma
control_interconnect
memory_interconnect
accelerator_control_cdc
dma_control_cdc
mm2s_cdc
s2mm_cdc
0
5000
10000
15000
20000
25000
30000
35000
Primitive count

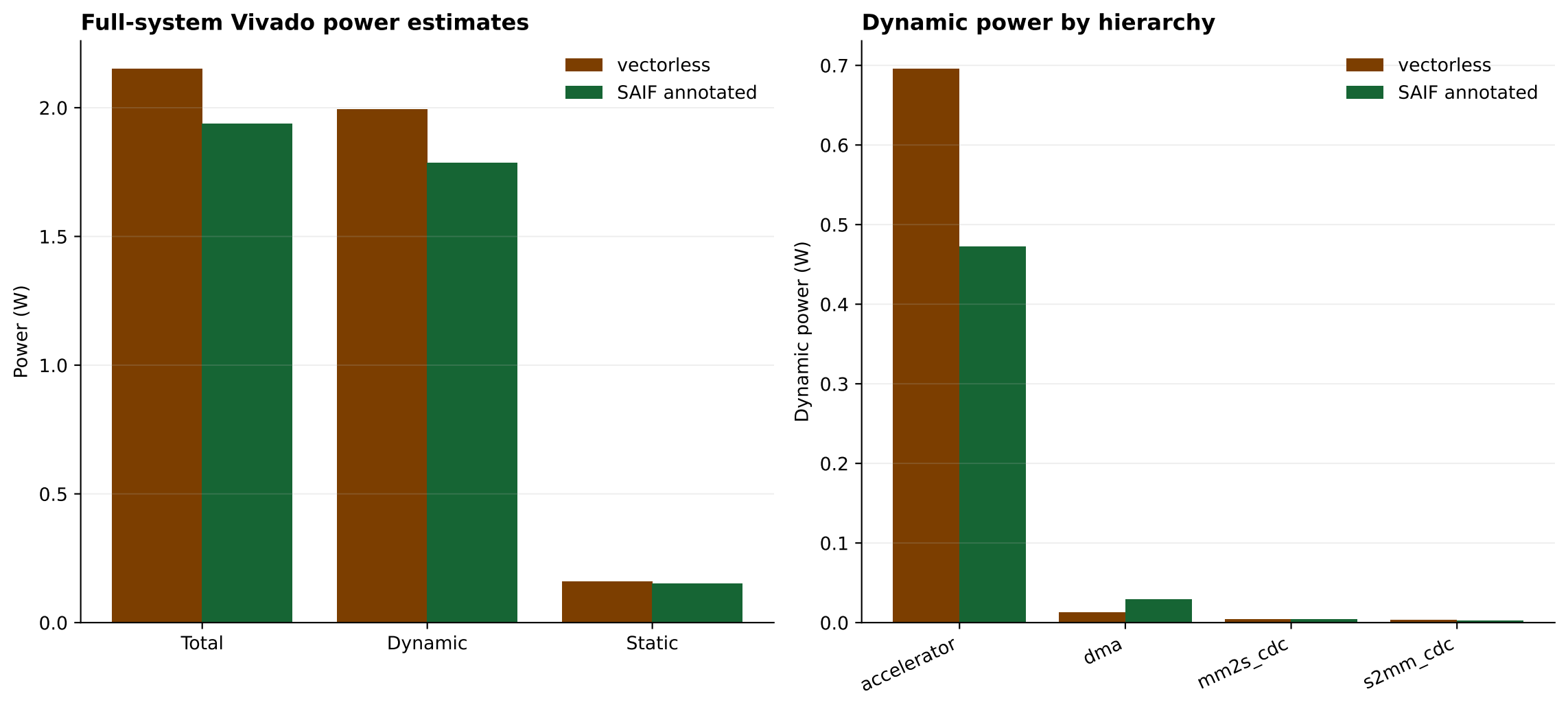

Flip-flops
accelerator
dma
control_interconnect
memory_interconnect
accelerator_control_cdc
dma_control_cdc
mm2s_cdc
s2mm_cdc
0
10000
20000
30000
40000
Primitive count


DSP48 blocks
accelerator
dma
control_interconnect
memory_interconnect
accelerator_control_cdc
dma_control_cdc
mm2s_cdc
s2mm_cdc
0
20
40
60
80
100
Primitive count


RAMB36 blocks
accelerator
dma
control_interconnect
memory_interconnect
accelerator_control_cdc
dma_control_cdc
mm2s_cdc
s2mm_cdc
0
5
10
15
20
25
Primitive count


Vivado DRC advisory classification
DPIP-1: Input pipelining
DPOP-1: PREG Output pipelining
DPOP-2: MREG Output pipelining
PDCN-1569: LUT equation term check
REQP-1839: RAMB36 async control check
REQP-1840: RAMB18 async control check
RTSTAT-10: No routable loads
10^0
10^1
10^2
Reported checks (log scale)


Vivado clock-domain crossing classification
Reported crossings
0
20
40
60
80
100
120
140
CDC-3 Info
1-bit synchronized with ASYNC_REG property
CDC-15 Warning
Clock enable controlled CDC structure detected


Full-system Vivado power estimates
vectorless
SAIF annotated
Power (W)
0.0
0.5
1.0
1.5
2.0
Total
Dynamic
Static
Dynamic power by hierarchy
vectorless
SAIF annotated
Dynamic power (W)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
accelerator
dma
mm2s_cdc
s2mm_cdc

Power Evidence and Confidence Boundary

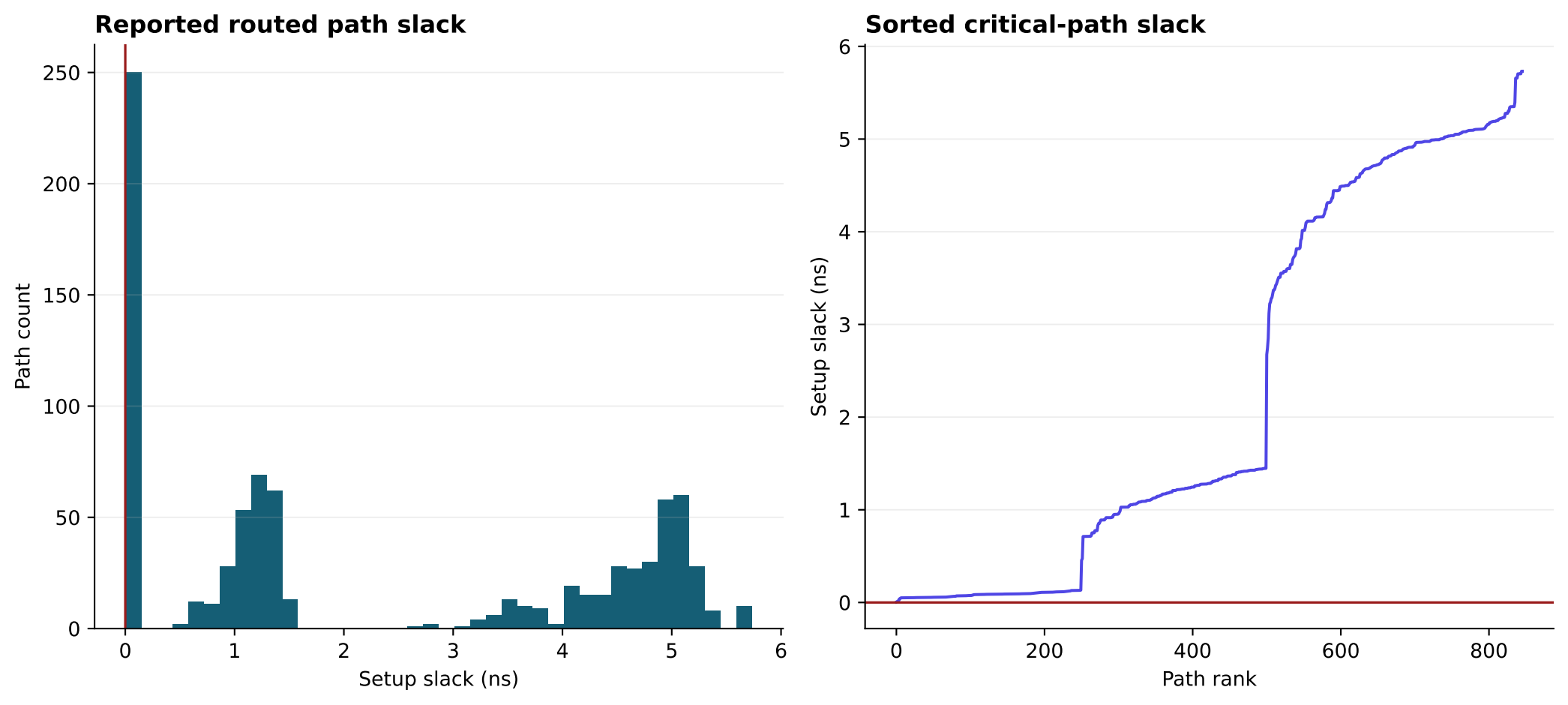
Vectorless: total 2.152 W, dynamic 1.993 W, confidence Medium

SAIF: total 1.937 W, dynamic 1.785 W, confidence Medium

SAIF implementation nets matched: 7%   (9164/123250)

Captured tests do not initialize every memory location; Vivado reports X-state warnings for untouched storage.

These are tool estimates for the routed design, not measurements from a PYNQ-Z2 power rail.


Reported routed path slack
Path count
Setup slack (ns)
0
50
100
150
200
250
0
1
2
3
4
5
6
Sorted critical-path slack
Setup slack (ns)
Path rank
0
1
2
3
4
5
6
0
200
400
600
800